

# E16866250\_3

The equation  $x^5 - x + y^3 + 4y + 1 = 0$  has no integer solutions

Size  $h = 51$ , length  $l = 12$

**Theorem 1.** *The equation*

$$x^5 - x + y^3 + 4y + 1 = 0 \tag{1}$$

*has no integer solutions.*

The proof uses the quartic Hilbert reciprocity law over the field  $K = \mathbb{Q}(i)$ . To a hypothetical integer solution we attach the values of 82 auxiliary polynomials  $H_0, \dots, H_{81}$  with coefficients in  $\mathbb{Z}[i]$ , constants  $c_0, \dots, c_{81} \in \mathbb{Z}[i]$ , and a sum  $B$  of quartic Hilbert symbols of these values, recorded by a graph with 676 weighted edges. By the reciprocity law, the local contributions  $B_p$  of all primes  $p$  add up to 0 modulo 4. We show that  $B_p = 0$  for  $p \notin S$ , compute  $B_p$  for the primes  $p \in S$ , and obtain a contradiction. Two standard facts from class field theory are quoted without proof: the explicit formula for the quartic Hilbert symbol at the places not above 2, and the Hilbert reciprocity law. Apart from these, the proof uses polynomial identities, which can be checked by expanding both sides, and finite computations with residues.

## 1 Quartic Hilbert symbols over $\mathbb{Q}(i)$

**The field.** Let  $K = \mathbb{Q}(i)$ , where  $i^2 = -1$ . Its ring of integers is  $\mathbb{Z}[i]$ , and every element of  $K$  can be written uniquely as  $A + Bi$  with  $A, B \in \mathbb{Q}$ . We write

$$\langle A, B \rangle = A + Bi,$$

also when  $A$  and  $B$  are polynomials in  $x$  and  $y$ . Then  $\langle A, B \rangle \langle C, D \rangle = \langle AC - BD, AD + BC \rangle$ , and the norm is  $N(\langle A, B \rangle) = A^2 + B^2$ . In particular, a value  $\langle A, B \rangle$  with  $A, B \in \mathbb{Z}$  is zero if and only if  $A = B = 0$ . The units of  $\mathbb{Z}[i]$  are  $\pm 1$  and  $\pm i$ .

**Places.** The finite places of  $K$  correspond to the non-zero prime ideals of  $\mathbb{Z}[i]$ . A prime  $p \equiv 3 \pmod{4}$  is inert: there is one place above  $p$ , with uniformizer  $p$  and residue field  $\mathbb{Z}[i]/p\mathbb{Z}[i]$  with  $p^2$  elements. If  $p \equiv 1 \pmod{4}$ , the polynomial  $t^2 + 1$  has two roots  $r$  modulo  $p$ , and for each of them there is a place  $v_r$  above  $p$ , at which  $i$  is the root of  $t^2 + 1$  in  $\mathbb{Z}_p$  congruent to  $r$  modulo  $p$ ; the completion is  $\mathbb{Q}_p$ , the uniformizer is  $p$ , the residue field is  $\mathbb{F}_p$ , and  $i$  reduces to  $r$ . The prime 2 is ramified:  $\pi = 1 + i$  generates the unique prime ideal above 2,  $\pi^2 = 2i$ , and  $\mathbb{Z}[i]/\pi\mathbb{Z}[i] = \mathbb{F}_2$ . The only infinite place of  $K$  is complex. For a finite place  $v$  we write  $K_v$  for the completion and  $v(a)$  for the normalized valuation, and we call  $a$  a *unit* at  $v$  if  $v(a) = 0$ .

**Hilbert symbols.** For a place  $v$  of  $K$  and  $a, b \in K_v^\times$ , the quartic Hilbert symbol  $(a, b)_v$  is a fourth root of unity, see [1, Chapter V, §3]. We write  $(a, b)_v = i^{\beta_v(a, b)}$  with  $\beta_v(a, b) \in \mathbb{Z}/4\mathbb{Z}$ , and we use the following three facts [1, Chapter V, §3 and Chapter VI, §8].

(H1)  $\beta_v(a, bc) = \beta_v(a, b) + \beta_v(a, c)$ ,  $\beta_v(a, b) = -\beta_v(b, a)$ , and  $\beta_v(a, c^4) = 0$ . In particular,  $\beta_v(a, b)$  depends only on the classes of  $a$  and  $b$  in  $K_v^\times/K_v^{\times 4}$ .

(H2) Let  $v$  be a finite place not above 2, let  $q$  be the number of elements of its residue field, and let  $\varpi$  be a uniformizer at  $v$ . Write  $a = \varpi^e u$  and  $b = \varpi^f w$  with units  $u$  and  $w$ , and define  $\chi(u) \in \mathbb{Z}/4\mathbb{Z}$  by  $\bar{u}^{(q-1)/4} = \bar{i}^{\chi(u)}$ , where the bar denotes the reduction to the residue field. Then  $\beta_v(a, b) = e\chi(w) - f\chi(u) + ef\chi(-1)$ , and  $\chi(-1) = (q-1)/2 \pmod{4}$ . In particular,  $\beta_v(a, b) = 0$  if  $a$  and  $b$  are both units.

(H3) (Hilbert reciprocity.) For  $a, b \in K^\times$ ,  $\beta_v(a, b) = 0$  for all but finitely many places  $v$ , and  $\sum_v \beta_v(a, b) = 0$ , where the sum is over all places of  $K$ . At the complex place,  $\beta_v(a, b) = 0$ .

The value of  $\chi(-1)$  follows from  $-1 = \bar{i}^2$  and  $(-1)^{(q-1)/4} = \bar{i}^{(q-1)/2}$ . Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values  $\beta_v$  change sign, and the proof below is the same.

**Classes at the places not above 2.** Let  $p \neq 2$  be a prime and  $v$  a place above  $p$ . For  $a = p^e u \in K_v^\times$  with a unit  $u$ , put  $c_v(a) = (e \pmod{4}, \chi(u)) \in (\mathbb{Z}/4\mathbb{Z})^2$ . By (H2) and (H1),  $c_v$  is additive,  $c_v(a)$  determines the class of  $a$  modulo fourth powers, and  $\beta_v(a, b) = e\chi(w) - f\chi(u) + \varepsilon_v ef$  for  $c_v(a) = (e, \chi(u))$  and  $c_v(b) = (f, \chi(w))$ , where  $\varepsilon_v = \chi(-1)$ . We put  $c_p(a) = c_v(a) \in (\mathbb{Z}/4\mathbb{Z})^2$  if  $p \equiv 3 \pmod{4}$ ; then  $q = p^2 \equiv 1 \pmod{8}$  and  $\varepsilon_v = 0$ . If  $p \equiv 1 \pmod{4}$ , we put  $c_p(a) = (c_{v_r}(a), c_{v_{r'}}(a)) \in (\mathbb{Z}/4\mathbb{Z})^4$ , where  $r < r'$  are the roots of  $t^2 + 1$  in  $\{1, \dots, p-1\}$ ; here  $q = p$ , and  $\varepsilon_v = 0$  if  $p \equiv 1 \pmod{8}$  and  $\varepsilon_v = 2$  if  $p \equiv 5 \pmod{8}$ . Accordingly,  $\beta_p$  is the sum of the forms  $\beta_v$  over the places  $v$  above  $p$ . For  $p \equiv 3 \pmod{4}$  and a rational integer  $u$  prime to  $p$ , we have  $\bar{u} \in \mathbb{F}_p$ , so that  $\bar{u}^{(p^2-1)/4} = (\bar{u}^{p-1})^{(p+1)/4} = 1$  and  $\chi(u) = 0$ .

### The place above 2.

**Lemma 2.** *Let  $v$  be the place above 2. Every non-zero element of  $K_v$  is equal to  $\pi^a i^b (1 + 2i)^{c5^d}$  times a fourth power, for unique  $a, b, c, d \in \mathbb{Z}/4\mathbb{Z}$ , and every unit congruent to 1 modulo  $\pi^7$  is a fourth power. Put  $c_2(A) = (a, b, c, d) \in (\mathbb{Z}/4\mathbb{Z})^4$ . Then  $c_2$  is additive, and  $\beta_v(A, B) = c_2(A)^\top M_2 c_2(B)$ , where*

$$M_2 = \begin{pmatrix} 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 2 \\ 1 & 3 & 2 & 0 \\ 3 & 2 & 0 & 0 \end{pmatrix} \pmod{4}.$$

*Proof.* For  $\alpha \in \mathbb{Z}[i]$ , the polynomial  $g(s) = s - 3(1-i)s^2 - 8is^3 + 4(1+i)s^4 + \alpha$  satisfies  $g'(s) \equiv 1 \pmod{\pi}$ , so by Hensel's lemma it has a root  $s$  in the completion of  $\mathbb{Z}[i]$ . Since  $(1 + \pi^3 s)^4 = 1 - \pi^7(s - 3(1-i)s^2 - 8is^3 + 4(1+i)s^4)$ , as one checks using  $\pi^5 = -4 - 4i$ ,  $\pi^6 = -8i$  and  $\pi^7 = 8 - 8i$ , we obtain  $(1 + \pi^3 s)^4 = 1 + \pi^7 \alpha$ . Hence every unit congruent to 1 modulo  $\pi^7$  is a fourth power in  $K_v$ , and the class of a unit modulo fourth powers depends only on its residue modulo  $\pi^7$ . The ring  $\mathbb{Z}[i]/\pi^7 \mathbb{Z}[i]$  has  $2^7$  elements, 64 of which are units, and a direct enumeration shows that the 64 products  $i^b (1 + 2i)^{c5^d}$  with  $b, c, d \in \{0, 1, 2, 3\}$  are pairwise distinct modulo  $\pi^7$ . Hence they represent all units modulo  $\pi^7$ , the group of these units has exponent 4, and the fourth power of every unit is congruent to 1 modulo  $\pi^7$ . Therefore two units are in the same class modulo fourth powers if and only if they are congruent modulo  $\pi^7$ , and the 64 products represent the 64 classes of units. Since  $\pi^4$  is a fourth power, the exponent of  $\pi$  is the valuation modulo 4, and the first statement follows.

By (H1), it remains to compute  $\beta_v$  on the 16 ordered pairs of the elements  $\pi, i, 1 + 2i$  and 5. Their norms are 2, 1, 5 and 25, hence, for each such pair, all the symbols at the places not above 2 and 5 vanish by (H2), and, by (H3),  $\beta_v$  at the place above 2 is minus the sum of the values at the two places above 5. These places correspond to  $\bar{i} = 2$  and  $\bar{i} = 3$  in  $\mathbb{F}_5$ ; there 5 is a uniformizer,  $q = 5$ ,  $\chi(u)$  is defined by  $\bar{u} = \bar{i}^{\chi(u)}$ , and  $\chi(-1) = 2$ . The classes  $(e, \chi)$  of the four

elements at these places are

|          | $\bar{i} = 2$ | $\bar{i} = 3$ |
|----------|---------------|---------------|
| $\pi$    | $(0, 3)$      | $(0, 2)$      |
| $i$      | $(0, 1)$      | $(0, 1)$      |
| $1 + 2i$ | $(1, 3)$      | $(0, 3)$      |
| $5$      | $(1, 0)$      | $(1, 0)$      |

where, for  $1 + 2i$  at the place with  $\bar{i} = 2$ , the unit part is  $(1 + 2i)/5 \equiv (1 + 2 \cdot 7)/5 = 3 \pmod{5}$ , using  $i \equiv 7 \pmod{25}$ . Formula (H2) and the sum over these two places give  $M_2$ .  $\square$

We write  $\beta_2$  for the form of Lemma 2. Since  $2 = i^3\pi^2$ , we have  $c_2(2) = (2, 3, 0, 0)$ ; also  $c_2(-1) = (0, 2, 0, 0)$  and  $c_2(5) = (0, 0, 0, 1)$ .

**Residue classes.** The local computations use the following description of the classes of the values of a polynomial on a residue class.

**Lemma 3.** *Let  $p$  be a prime,  $k \geq 1$ , and let  $z \in \mathbb{Z}[i]$  be non-zero with  $z \equiv \zeta \pmod{p^k}$ , where  $\zeta = \langle A, B \rangle$  with  $0 \leq A, B < p^k$ . Then  $c_p(z) \in b + D$  for the vector  $b$  and the subgroup  $D$  given as follows.*

- (i)  $p \equiv 3 \pmod{4}$ . If  $\zeta \neq 0$ , then  $b = c_p(\zeta)$  and  $D = 0$ . If  $\zeta = 0$ , then  $b = 0$ , and  $D$  is  $(\mathbb{Z}/4\mathbb{Z})^2$ , or the span of  $(1, 0)$  if  $z \in \mathbb{Z}$ .
- (ii)  $p \equiv 1 \pmod{4}$ . For each root  $r$ , let  $\rho$  be the root of  $t^2 + 1$  modulo  $p^k$  congruent to  $r$ . If  $A + B\rho \not\equiv 0 \pmod{p^k}$ , the two coordinates of  $c_{v_r}(z)$  are those of  $c_{v_r}(\zeta)$ ; otherwise these two coordinates are arbitrary.
- (iii)  $p = 2$ ,  $z \notin \mathbb{Z}$ . If  $\zeta \neq 0$ , let  $a = v(\zeta) < 2k$  be its valuation at  $\pi$  and  $m = 2k - a$ . Then  $b = c_2(\zeta)$ , and  $D = 0$  if  $m \geq 7$ , while  $D = D_m$  if  $m \leq 6$ , where  $D_1$  is spanned by  $(0, 1, 0, 0)$ ,  $(0, 0, 1, 0)$ , and  $(0, 0, 0, 1)$ ;  $D_2$  is spanned by  $(0, 2, 0, 0)$ ,  $(0, 0, 1, 0)$ , and  $(0, 0, 0, 1)$ ;  $D_3$  is spanned by  $(0, 2, 1, 0)$ ,  $(0, 0, 2, 0)$ , and  $(0, 0, 0, 1)$ ;  $D_4$  is spanned by  $(0, 0, 2, 0)$  and  $(0, 0, 0, 1)$ ;  $D_5$  is spanned by  $(0, 0, 2, 0)$  and  $(0, 0, 0, 2)$ ;  $D_6$  is spanned by  $(0, 0, 0, 2)$ . If  $\zeta = 0$ , then  $b = 0$  and  $D = (\mathbb{Z}/4\mathbb{Z})^4$ .
- (iv)  $p = 2$ ,  $z \in \mathbb{Z}$ , so that  $B = 0$ . If  $2^k \nmid A$ , let  $j = v_2(A)$  and  $u = A/2^j$ . Then  $b = c_2(A)$  and  $D = 0$  if  $k - j \geq 4$ , while  $b = j c_2(2) + c_2(u)$  and  $D = R_{k-j}$  if  $k - j \leq 3$ , where  $R_1$  is spanned by  $(0, 2, 0, 0)$  and  $(0, 0, 0, 1)$ ;  $R_2$  is spanned by  $(0, 0, 0, 1)$ ;  $R_3$  is spanned by  $(0, 0, 0, 2)$ . If  $2^k \mid A$ , then  $b = 0$  and  $D$  is spanned by  $c_2(2)$ ,  $c_2(-1)$  and  $c_2(5)$ .

*Proof.* (i) If  $\zeta \neq 0$ , then  $e = v(z) = v(\zeta) < k$  and  $z/p^e \equiv \zeta/p^e \pmod{p}$ , so that  $\chi$  is determined. If  $z \in \mathbb{Z}$ , its unit part is a rational integer, and  $\chi = 0$ . (ii) At  $v_r$ ,  $z$  and  $\zeta$  are the  $p$ -adic numbers  $A' + B'i$  and  $A + Bi$  with  $i \equiv \rho \pmod{p^k}$ , and the argument of (i) applies. (iii) If  $\zeta \neq 0$ , then  $v(z) = a$ , and  $z/\pi^a \equiv \zeta/\pi^a \pmod{\pi^m}$ . By Lemma 2, the class of a unit depends only on its residue modulo  $\pi^7$ ; the units congruent to 1 modulo  $\pi^m$  form a subgroup, and a direct enumeration of their residues modulo  $\pi^7$  shows that their classes form  $D_m$ . (iv) If  $2^k \nmid A$ , then  $z = 2^j u'$  with a rational unit  $u' \equiv u \pmod{2^{k-j}}$ , and  $u'/u$  is a rational unit congruent to 1 modulo  $2^{k-j}$ . A rational unit congruent to 1 modulo 16 is congruent to 1 modulo  $\pi^8$ , hence a fourth power, and the odd residues modulo 16 are  $\pm 5^s$ ; hence the classes of the rational units congruent to 1 modulo  $2^{k-j}$  form  $R_{k-j}$ , and this group is 0 if  $k - j \geq 4$ . If  $2^k \mid A$ , then also the valuation  $j$  is unknown, and every non-zero rational number is  $2^j(\pm 5^s)$  times a fourth power.  $\square$

**Lemma 4.** *Let  $p$  be a prime, and consider vertices  $i$  with vectors  $b_i$ , subgroups  $D_i$  and vectors  $\gamma_i$ , and a set of edges  $(i, j)$  with  $i < j$  and weights  $w_{ij} \in \{1, 2, 3\}$ . Put  $\rho_{ij} = w_{ij}$  and  $\rho_{ji} = -w_{ij}$*

for the edges, and  $\sigma_i = \gamma_i + \sum_j \rho_{ij} b_j$ . Suppose that  $\beta_p(d, \sigma_i) = 0$  for every  $i$  and every  $d \in D_i$ , and that  $\beta_p(d, d') = 0$  for every edge  $(i, j)$  and all  $d \in D_i, d' \in D_j$ . Then

$$\sum_{(i,j)} w_{ij} \beta_p(c_i, c_j) + \sum_i \beta_p(c_i, \gamma_i) = \sum_{(i,j)} w_{ij} \beta_p(b_i, b_j) + \sum_i \beta_p(b_i, \gamma_i)$$

whenever  $c_i \in b_i + D_i$  for every  $i$ . By bilinearity, it suffices to check the hypotheses for  $d$  and  $d'$  in generating sets of  $D_i$  and  $D_j$ .

*Proof.* Write  $c_i = b_i + d_i$  with  $d_i \in D_i$ . As  $\beta_p$  is bilinear and  $\beta_p(u, v) = -\beta_p(v, u)$ , the left side equals the right side plus  $\sum_i \beta_p(d_i, \sigma_i) + \sum_{(i,j)} w_{ij} \beta_p(d_i, d_j)$ , and both sums vanish.  $\square$

## 2 The graph

Put  $F(x, y) = x^5 - x + y^3 + 4y + 1$ , so that (1) is the equation  $F(x, y) = 0$ . The certificate consists of 82 polynomials  $H_i$  with coefficients in  $\mathbb{Z}[i]$  and constants  $c_i \in \mathbb{Z}[i]$ , listed in Table 1, and of 676 edges  $(i, j)$  with  $i < j$  and weights  $w_{ij} \in \{1, 2, 3\}$ , listed below. For an edge  $(i, j)$  we put  $\rho_{ij} = w_{ij}$  and  $\rho_{ji} = -w_{ij} \bmod 4$ , and  $\rho_{ij} = 0$  if  $i$  and  $j$  are not joined by an edge. The edges  $(i, j; w_{ij})$  are (0, 9; 2), (0, 13; 2), (0, 16; 2), (0, 23; 1), (0, 31; 3), (0, 39; 1), (0, 41; 1), (0, 55; 1), (0, 67; 1), (0, 69; 1), (0, 73; 2), (0, 79; 2), (1, 14; 1), (1, 18; 3), (1, 22; 2), (1, 25; 2), (1, 26; 1), (1, 29; 2), (1, 31; 3), (1, 35; 1), (1, 37; 2), (1, 41; 3), (1, 48; 3), (1, 55; 3), (1, 61; 1), (1, 62; 1), (1, 66; 3), (1, 75; 1), (1, 77; 3), (1, 80; 2), (2, 8; 2), (2, 12; 2), (2, 17; 1), (2, 18; 3), (2, 20; 1), (2, 21; 3), (2, 25; 3), (2, 26; 3), (2, 33; 2), (2, 34; 2), (2, 35; 2), (2, 38; 2), (2, 39; 1), (2, 45; 3), (2, 52; 1), (2, 62; 2), (2, 69; 1), (2, 71; 1), (2, 73; 2), (2, 75; 3), (3, 7; 3), (3, 9; 1), (3, 16; 1), (3, 25; 1), (3, 32; 3), (3, 34; 1), (3, 38; 3), (3, 40; 1), (3, 41; 2), (3, 54; 1), (3, 57; 1), (3, 58; 3), (3, 65; 3), (3, 68; 2), (3, 73; 1), (3, 74; 2), (3, 80; 3), (4, 6; 3), (4, 7; 2), (4, 16; 3), (4, 20; 3), (4, 25; 1), (4, 28; 2), (4, 29; 3), (4, 31; 1), (4, 35; 2), (4, 38; 2), (4, 42; 3), (4, 44; 3), (4, 57; 3), (4, 59; 3), (4, 67; 2), (4, 73; 2), (4, 75; 2), (4, 81; 2), (5, 9; 2), (5, 12; 1), (5, 13; 2), (5, 14; 3), (5, 16; 3), (5, 17; 3), (5, 25; 3), (5, 26; 1), (5, 27; 2), (5, 35; 3), (5, 36; 2), (5, 39; 2), (5, 63; 3), (5, 66; 3), (5, 67; 3), (5, 72; 3), (6, 10; 1), (6, 11; 2), (6, 14; 1), (6, 20; 3), (6, 25; 1), (6, 28; 3), (6, 30; 1), (6, 32; 1), (6, 33; 2), (6, 35; 2), (6, 40; 3), (6, 41; 2), (6, 42; 3), (6, 48; 1), (6, 51; 2), (6, 52; 3), (6, 53; 2), (6, 56; 2), (6, 64; 3), (6, 66; 2), (6, 80; 1), (7, 11; 3), (7, 28; 3), (7, 29; 1), (7, 32; 2), (7, 35; 3), (7, 38; 1), (7, 39; 2), (7, 40; 3), (7, 41; 2), (7, 42; 2), (7, 43; 1), (7, 48; 1), (7, 50; 2), (7, 53; 3), (7, 55; 3), (7, 57; 1), (7, 62; 2), (7, 63; 3), (7, 65; 2), (7, 75; 1), (7, 77; 3), (7, 78; 3), (8, 11; 3), (8, 15; 2), (8, 32; 3), (8, 40; 1), (8, 41; 1), (8, 42; 2), (8, 48; 2), (8, 49; 3), (8, 50; 3), (8, 57; 3), (8, 59; 2), (8, 62; 3), (8, 70; 2), (8, 79; 2), (8, 80; 2), (9, 10; 1), (9, 16; 3), (9, 18; 2), (9, 21; 3), (9, 32; 1), (9, 35; 1), (9, 37; 3), (9, 38; 2), (9, 41; 2), (9, 46; 3), (9, 48; 2), (9, 49; 2), (9, 52; 2), (9, 58; 1), (9, 64; 2), (9, 72; 2), (10, 14; 2), (10, 15; 1), (10, 20; 2), (10, 30; 2), (10, 36; 1), (10, 41; 2), (10, 45; 2), (10, 51; 3), (10, 61; 1), (10, 62; 1), (10, 69; 3), (10, 70; 1), (10, 72; 2), (10, 77; 3), (10, 78; 2), (10, 80; 3), (11, 14; 3), (11, 15; 1), (11, 41; 1), (11, 45; 1), (11, 54; 3), (11, 60; 3), (11, 69; 2), (11, 70; 1), (11, 76; 1), (12, 16; 2), (12, 18; 1), (12, 25; 3), (12, 31; 1), (12, 34; 2), (12, 35; 2), (12, 38; 3), (12, 41; 2), (12, 48; 3), (12, 49; 3), (12, 61; 1), (12, 63; 1), (12, 64; 1), (12, 66; 2), (12, 72; 1), (12, 77; 1), (13, 19; 2), (13, 25; 2), (13, 41; 2), (13, 63; 2), (13, 69; 2), (13, 71; 2), (14, 26; 1), (14, 35; 3), (14, 41; 1), (14, 45; 1), (14, 46; 3), (14, 52; 3), (14, 60; 1), (14, 64; 1), (14, 66; 1), (14, 69; 3), (15, 24; 1), (15, 29; 2), (15, 30; 2), (15, 34; 1), (15, 45; 3), (15, 46; 3), (15, 53; 2), (15, 60; 3), (15, 62; 1), (15, 63; 1), (15, 65; 2), (15, 69; 1), (15, 73; 3), (16, 20; 3), (16, 21; 1), (16, 23; 3), (16, 25; 1), (16, 36; 3), (16, 37; 3), (16, 49; 3), (16, 58; 1), (16, 59; 2), (16, 64; 1), (16, 65; 3), (16, 68; 1), (16, 69; 3), (16, 78; 1), (16, 79; 3), (17, 18; 3), (17, 29; 2), (17, 30; 1), (17, 45; 1), (17, 49; 1), (17, 50; 2), (17, 51; 1), (17, 53; 3), (17, 56; 1), (17, 58; 1), (17, 59; 1), (17, 64; 1), (17, 67; 3), (17, 81; 2), (18, 34; 1), (18, 35; 3), (18, 46; 1), (18, 47; 2), (18, 53; 1), (18, 58; 3), (18, 64; 3), (18, 65; 2), (18, 71; 2), (18, 75; 3), (18, 76; 1), (18, 77; 3), (19, 20; 2), (19, 25; 2), (19, 42; 2), (19, 58; 2), (19, 70; 2), (19, 73; 2),

(20, 22; 2), (20, 24; 1), (20, 25; 3), (20, 35; 3), (20, 36; 1), (20, 38; 1), (20, 39; 3), (20, 43; 3),  
 (20, 50; 2), (20, 51; 1), (20, 56; 1), (20, 59; 1), (20, 63; 2), (20, 65; 3), (20, 66; 3), (20, 68; 1),  
 (21, 23; 3), (21, 26; 1), (21, 28; 3), (21, 32; 3), (21, 39; 3), (21, 42; 1), (21, 45; 2), (21, 47; 1),  
 (21, 53; 2), (21, 58; 3), (21, 60; 1), (21, 64; 1), (21, 65; 2), (21, 66; 1), (21, 69; 2), (21, 77; 1),  
 (21, 80; 1), (22, 30; 2), (22, 35; 2), (22, 39; 2), (22, 68; 2), (22, 79; 2), (22, 80; 2), (23, 29; 1),  
 (23, 30; 2), (23, 37; 1), (23, 39; 2), (23, 44; 2), (23, 45; 1), (23, 49; 2), (23, 57; 3), (23, 59; 1),  
 (23, 60; 2), (23, 66; 3), (24, 26; 2), (24, 34; 3), (24, 36; 1), (24, 37; 2), (24, 39; 1), (24, 44; 3),  
 (24, 55; 1), (24, 59; 2), (24, 62; 3), (24, 68; 1), (24, 70; 1), (24, 73; 1), (24, 76; 3), (24, 77; 3),  
 (25, 31; 1), (25, 32; 3), (25, 35; 3), (25, 39; 3), (25, 45; 3), (25, 47; 3), (25, 52; 2), (25, 54; 2),  
 (25, 61; 3), (25, 64; 1), (25, 69; 3), (25, 71; 2), (25, 77; 1), (25, 78; 1), (25, 80; 2), (26, 27; 3),  
 (26, 28; 3), (26, 29; 3), (26, 31; 3), (26, 34; 2), (26, 35; 3), (26, 36; 2), (26, 41; 3), (26, 43; 3),  
 (26, 44; 1), (26, 46; 3), (26, 54; 3), (26, 55; 3), (26, 58; 1), (26, 59; 3), (26, 62; 3), (26, 64; 2),  
 (26, 65; 2), (26, 66; 1), (26, 67; 1), (26, 68; 3), (26, 69; 2), (26, 71; 1), (26, 73; 3), (26, 79; 3),  
 (27, 29; 2), (27, 31; 2), (27, 38; 1), (27, 43; 3), (27, 45; 2), (27, 46; 1), (27, 50; 2), (27, 55; 2),  
 (27, 56; 3), (27, 59; 1), (27, 62; 2), (27, 69; 1), (27, 71; 1), (27, 75; 3), (27, 76; 1), (27, 79; 1),  
 (27, 80; 3), (28, 31; 2), (28, 32; 1), (28, 46; 1), (28, 50; 1), (28, 51; 1), (28, 54; 3), (28, 62; 1),  
 (28, 64; 3), (28, 67; 3), (28, 71; 2), (28, 72; 1), (29, 32; 2), (29, 41; 1), (29, 45; 2), (29, 47; 3),  
 (29, 50; 2), (29, 55; 1), (29, 58; 3), (29, 61; 1), (29, 65; 1), (29, 71; 2), (29, 72; 3), (30, 31; 2),  
 (30, 33; 2), (30, 41; 2), (30, 46; 1), (30, 47; 2), (30, 50; 3), (30, 52; 1), (30, 62; 3), (30, 63; 2),  
 (30, 73; 2), (30, 78; 1), (30, 80; 3), (31, 37; 2), (31, 41; 1), (31, 42; 2), (31, 53; 1), (31, 63; 2),  
 (31, 66; 3), (31, 70; 2), (31, 75; 3), (31, 78; 3), (32, 38; 2), (32, 44; 2), (32, 49; 3), (32, 50; 1),  
 (32, 62; 2), (32, 65; 1), (32, 68; 2), (32, 73; 3), (32, 75; 1), (32, 78; 1), (32, 80; 3), (33, 36; 2),  
 (33, 53; 2), (33, 64; 2), (33, 66; 2), (33, 69; 2), (34, 38; 1), (34, 42; 1), (34, 44; 1), (34, 48; 1),  
 (34, 51; 2), (34, 53; 2), (34, 55; 1), (34, 56; 2), (34, 60; 1), (34, 62; 1), (34, 64; 1), (34, 77; 2),  
 (34, 78; 3), (34, 81; 2), (35, 37; 2), (35, 44; 2), (35, 45; 3), (35, 46; 1), (35, 48; 1), (35, 53; 2),  
 (35, 60; 3), (35, 62; 2), (35, 64; 2), (35, 70; 3), (35, 77; 3), (35, 79; 2), (36, 37; 1), (36, 45; 1),  
 (36, 46; 3), (36, 48; 3), (36, 49; 2), (36, 53; 3), (36, 57; 3), (36, 59; 2), (36, 61; 1), (36, 62; 3),  
 (36, 64; 3), (36, 65; 1), (36, 66; 1), (36, 68; 1), (36, 69; 1), (36, 73; 1), (36, 79; 2), (37, 42; 1),  
 (37, 45; 1), (37, 47; 2), (37, 49; 3), (37, 50; 2), (37, 53; 2), (37, 62; 3), (37, 64; 2), (37, 66; 3),  
 (37, 76; 3), (37, 78; 2), (38, 44; 3), (38, 48; 1), (38, 49; 1), (38, 51; 2), (38, 54; 2), (38, 56; 3),  
 (38, 60; 2), (38, 63; 1), (38, 64; 2), (38, 65; 2), (38, 66; 2), (38, 67; 1), (38, 74; 3), (38, 75; 3),  
 (38, 79; 1), (38, 80; 1), (38, 81; 2), (39, 44; 2), (39, 49; 2), (39, 50; 1), (39, 52; 2), (39, 53; 3),  
 (39, 64; 3), (39, 79; 2), (39, 80; 3), (40, 44; 1), (40, 62; 3), (41, 45; 3), (41, 59; 1), (41, 60; 1),  
 (41, 61; 3), (41, 66; 2), (41, 70; 3), (41, 77; 3), (41, 80; 1), (42, 45; 1), (42, 50; 3), (42, 56; 2),  
 (42, 64; 2), (42, 70; 1), (42, 73; 1), (42, 77; 3), (43, 52; 3), (43, 78; 3), (44, 50; 1), (44, 55; 3),  
 (44, 56; 2), (44, 59; 3), (44, 76; 2), (45, 48; 3), (45, 50; 1), (45, 51; 2), (45, 62; 1), (45, 65; 2),  
 (45, 66; 2), (45, 76; 1), (45, 77; 1), (46, 49; 3), (46, 54; 1), (46, 57; 1), (46, 64; 2), (46, 70; 2),  
 (46, 78; 1), (47, 63; 1), (48, 50; 2), (48, 60; 3), (48, 65; 2), (48, 66; 2), (48, 70; 1), (48, 79; 1),  
 (48, 80; 2), (49, 51; 2), (49, 58; 2), (49, 69; 1), (49, 74; 1), (49, 78; 3), (50, 55; 1), (50, 56; 2),  
 (50, 66; 1), (50, 70; 3), (50, 73; 3), (51, 58; 2), (51, 59; 3), (51, 64; 2), (51, 67; 2), (51, 74; 3),  
 (51, 77; 3), (51, 79; 3), (51, 81; 2), (52, 71; 1), (53, 54; 2), (53, 60; 1), (53, 62; 2), (53, 65; 3),  
 (53, 71; 1), (53, 78; 1), (53, 80; 3), (54, 56; 3), (54, 60; 3), (54, 63; 1), (54, 71; 3), (55, 56; 3),  
 (55, 61; 3), (55, 73; 2), (55, 76; 3), (55, 77; 2), (55, 79; 2), (56, 60; 2), (56, 64; 2), (56, 77; 3),  
 (57, 59; 3), (57, 62; 1), (57, 63; 1), (57, 64; 3), (57, 70; 2), (57, 72; 3), (57, 81; 2), (58, 65; 1),  
 (58, 67; 1), (58, 68; 3), (58, 74; 3), (59, 66; 1), (59, 77; 2), (60, 64; 2), (60, 77; 1), (61, 62; 3),  
 (61, 64; 1), (62, 67; 2), (62, 68; 2), (62, 70; 2), (62, 72; 3), (62, 76; 3), (63, 72; 1), (64, 65; 1),  
 (64, 66; 3), (64, 72; 3), (64, 78; 2), (65, 67; 3), (65, 72; 1), (65, 76; 2), (66, 73; 1), (66, 74; 1),  
 (66, 81; 2), (67, 77; 3), (69, 70; 3), (71, 77; 3), (73, 77; 1), (74, 76; 3), (74, 77; 2), (77, 81; 2). Let

$(\xi, \eta)$  denote  $(x, y)$  or  $(y, x)$ . We call a vertex  $i$  *linear* if  $H_i = \varepsilon_i(\lambda_i \eta - s_i(\xi))$  for a unit  $\varepsilon_i$  of  $\mathbb{Z}[i]$ , an element  $\lambda_i \in \mathbb{Z}[i]$  whose norm is a power of 2, and a polynomial  $s_i$  with coefficients in  $\mathbb{Z}[i]$ ; the last column of Table 1 gives  $\lambda_i \eta = s_i(\xi)$  for the linear vertices, where  $\lambda_i = 1$  if no

factor is shown. For a linear vertex  $i$ , we put  $f_i(t) = \lambda_i^d F$  and  $g_{ij}(t) = \lambda_i^{d_j} H_j$  evaluated at  $\xi = t$ ,  $\eta = s_i(t)/\lambda_i$ , where  $d$  and  $d_j$  are the degrees of  $F$  and  $H_j$  in  $\eta$ ; these are polynomials in  $t$  with coefficients in  $\mathbb{Z}[i]$ . For every linear vertex  $i$ , Table 2 gives an integer  $D_i \geq 1$  and a polynomial  $w_i(t)$  with coefficients in  $\mathbb{Z}[i]$  such that

$$D_i^4 \lambda_i^{e_i} c_i \prod_j g_{ij}^{\rho_{ij}} - w_i^4 = f_i Q_i \quad (2)$$

for a polynomial  $Q_i(t)$  with coefficients in  $\mathbb{Z}[i]$ , where  $\rho_{ij} \in \{0, 1, 2, 3\}$ ; the table also gives an integer  $e_i \geq 0$  with  $e_i + \sum_j \rho_{ij} d_j \equiv 0 \pmod{4}$ , and  $e_i = 0$  if  $\lambda_i = 1$ ;  $Q_i$  is the quotient of the division of the left side by  $f_i$ , and the remainder is zero. For the edges  $(i, j)$  listed in Table 3, the table gives a linear endpoint  $k \in \{i, j\}$  and the factorization of the norm of the resultant  $R_{ij} = \text{Res}_t(f_k, g_{kl})$ , where  $\{k, l\} = \{i, j\}$ ; it is non-zero. These identities can be checked by expanding both sides and by polynomial division, using  $i^2 = -1$ .

### 3 Proof of Theorem 1

Suppose that  $(x, y) \in \mathbb{Z}^2$  is a solution of (1), so that  $F(x, y) = 0$ ; from now on the polynomials denote their values at  $(x, y)$ , which lie in  $\mathbb{Z}[i]$ .

**Step 1: the values are non-zero.** For  $i \in \{3, 4, 6, 7, 8, 9, 10, 11, 13, 14, 15, 18, 19, 28, 29, 31, 33, 39, 40, 41, 42, 43, 44, 45, 47, 50, 52, 61, 63, 68, 71, 73, 74, 78, 80\}$ , there is no residue pair  $(a, b)$  modulo 2 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{2}$ , as one checks for the  $2^2$  residue pairs; hence  $H_i \neq 0$ . For  $i \in \{1, 16, 20, 22, 27, 30, 34, 35, 36, 37, 38, 53, 55, 56, 57, 58, 59, 64, 67, 69, 70, 72, 76, 81\}$ , there is no residue pair  $(a, b)$  modulo 3 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{3}$ , as one checks for the  $3^2$  residue pairs; hence  $H_i \neq 0$ . For  $i \in \{12, 24, 25, 46, 54, 62, 65, 66, 75, 79\}$ , there is no residue pair  $(a, b)$  modulo 4 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{4}$ , as one checks for the  $4^2$  residue pairs; hence  $H_i \neq 0$ . For  $i \in \{17, 32, 51, 60\}$ , there is no residue pair  $(a, b)$  modulo 5 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{5}$ , as one checks for the  $5^2$  residue pairs; hence  $H_i \neq 0$ . For  $i \in \{0, 2, 5, 21, 23, 49\}$ , there is no residue pair  $(a, b)$  modulo 7 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{7}$ , as one checks for the  $7^2$  residue pairs; hence  $H_i \neq 0$ . For  $i \in \{77\}$ , there is no residue pair  $(a, b)$  modulo 11 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{11}$ , as one checks for the  $11^2$  residue pairs; hence  $H_i \neq 0$ . For  $i \in \{26\}$ , there is no residue pair  $(a, b)$  modulo 13 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{13}$ , as one checks for the  $13^2$  residue pairs; hence  $H_i \neq 0$ . For  $i \in \{48\}$ , there is no residue pair  $(a, b)$  modulo 29 with  $F(a, b) \equiv 0$  and  $H_i(a, b) \equiv 0 \pmod{29}$ , as one checks for the  $29^2$  residue pairs; hence  $H_i \neq 0$ . Also, all  $c_i$  are non-zero.

**Step 2: reciprocity.** For every place  $v$  of  $K$ , put

$$B_v = \sum_{(i,j)} w_{ij} \beta_v(H_i, H_j) + \sum_i \beta_v(H_i, c_i) \in \mathbb{Z}/4\mathbb{Z}, \quad (3)$$

the first sum over the edges, and, for a rational prime  $p$ , put  $B_p = \sum_{v|p} B_v$ . Applying (H3) to every pair  $(H_i, H_j)$  with an edge, multiplied by  $w_{ij}$ , and to every pair  $(H_i, c_i)$ , and adding, we find that  $B_v = 0$  for all but finitely many  $v$  and  $\sum_p B_p = 0$ , as the complex place contributes 0. By (H1) and the definition of  $c_p$ ,  $B_p = \sum_{(i,j)} w_{ij} \beta_p(c_p(H_i), c_p(H_j)) + \sum_i \beta_p(c_p(H_i), c_p(c_i))$ , where  $\beta_p$  is the form of Lemma 2 if  $p = 2$ .

**Step 3: the primes outside  $S$ .** Let  $S = \{2, 3, 5, 7, 11, 13, 17, 29, 37, 41, 53, 61\}$ . It contains 2 and the primes dividing the norms of the constants  $c_i$ , of the  $D_i$ , of the  $R_{ij}$ . Let  $p \notin S$ , and let  $v$  be a place above  $p$ , with residue field  $k_v$ ; then all these numbers are units at  $v$ .

First, no edge  $(i, j)$  has both values divisible by  $v$ . Indeed, suppose that  $v$  divides  $H_i$  and  $H_j$ . If  $R_{ij}$  is defined, let  $k$  be the linear endpoint used for it and  $\{k, l\} = \{i, j\}$ , and put  $t = \xi$ . As  $H_k = \varepsilon_k(\lambda_k \eta - s_k(\xi))$  and  $\lambda_k$  is a unit at  $v$ , we have  $\eta \equiv s_k(t)/\lambda_k$  modulo  $v$ , hence  $f_k(t) \equiv \lambda_k^d F(x, y) = 0$  and  $g_{kl}(t) \equiv \lambda_k^{d_l} H_l \equiv 0$  modulo  $v$ . The resultant can be written as  $R_{ij} = A f_k + C g_{kl}$  with polynomials  $A$  and  $C$  with coefficients in  $\mathbb{Z}[i]$ , so  $v$  divides  $R_{ij}$ , a contradiction. By (H2), a term of (3) in which all arguments are units vanishes. Now let  $v$  divide  $H_i$ . Collecting the terms of (3) which contain  $H_i$ , and using (H1) and  $\rho_{ji} = -w_{ij}$  for the edges  $(j, i)$  with  $j < i$ , we obtain  $\beta_v(H_i, \Pi_i)$  with  $\Pi_i = c_i \prod_j H_j^{\rho_{ij}}$ , which is a unit. We claim that  $D_i^4 \Pi_i \equiv W^4$  modulo  $v$  for some  $W \in K$  integral at  $v$ . If  $i$  is linear, put  $t = \xi$ ; then  $g_{ij}(t) \equiv \lambda_i^{d_j} H_j$  and  $f_i(t) \equiv 0$  modulo  $v$  as above, and (2) gives  $D_i^4 \lambda_i^{E_i} \Pi_i \equiv w_i(t)^4$  modulo  $v$ , where  $E_i = e_i + \sum_j \rho_{ij} d_j$  is divisible by 4; this gives the claim with  $W = w_i(t)/\lambda_i^{E_i/4}$ . Hence the reduction of  $\Pi_i$  is a non-zero fourth power in  $k_v$ ,  $\chi(\Pi_i) = 0$ , and  $\beta_v(H_i, \Pi_i) = v(H_i)\chi(\Pi_i) = 0$  by (H2). Hence  $B_v = 0$ , and  $B_p = 0$ .

**Step 4: the primes in  $S$ .** Let  $p \in S$ . Here 3 is inert, the places above 5 are  $v_2$  and  $v_3$ , with  $\varepsilon_v = 2$ , 7 is inert, 11 is inert, the places above 13 are  $v_5$  and  $v_8$ , with  $\varepsilon_v = 2$ , the places above 17 are  $v_4$  and  $v_{13}$ , with  $\varepsilon_v = 0$ , the places above 29 are  $v_{12}$  and  $v_{17}$ , with  $\varepsilon_v = 2$ , the places above 37 are  $v_6$  and  $v_{31}$ , with  $\varepsilon_v = 2$ , the places above 41 are  $v_9$  and  $v_{32}$ , with  $\varepsilon_v = 0$ , the places above 53 are  $v_{23}$  and  $v_{30}$ , with  $\varepsilon_v = 2$ , and the places above 61 are  $v_{11}$  and  $v_{50}$ , with  $\varepsilon_v = 2$ . Suppose that  $x \equiv a$  and  $y \equiv b \pmod{p^k}$ . Then  $H_i \equiv H_i(a, b) \pmod{p^k}$ , and Lemma 3, applied with  $\zeta$  the residue of  $H_i(a, b)$ , gives a vector  $b_i$  and a subgroup  $D_i$  with  $c_p(H_i) \in b_i + D_i$ ; when  $H_i$  has rational coefficients,  $H_i \in \mathbb{Z}$ , and we use this in (i) and (iv). With  $\gamma_i = c_p(c_i)$ , if the hypotheses of Lemma 4 hold, then  $B_p$  is determined by  $a$ ,  $b$  and  $k$ . Starting from the solutions of (1) modulo  $p$ , we refine a residue class  $x \equiv a$ ,  $y \equiv b \pmod{p^k}$  to the classes modulo  $p^{k+1}$  that contain solutions of (1) modulo  $p^{k+1}$ , until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes and the value of  $B_p$  in each; every integer solution lies in one of them. The result is  $B_2 \in \{1, 2\}$ ,  $B_3 \in \{2, 3\}$ ,  $B_5 = 2$ ,  $B_7 = 0$ ,  $B_{11} = 0$ ,  $B_{13} = 0$ ,  $B_{17} = 0$ ,  $B_{29} = 0$ ,  $B_{37} = 0$ ,  $B_{41} = 0$ ,  $B_{53} = 0$ ,  $B_{61} = 0$ .

**Step 5: contradiction.** By Steps 2 and 3,  $\sum_{p \in S} B_p = 0$ . But by Step 4,  $\sum_{p \in S} B_p \in \{1, 2, 3\}$  in  $\mathbb{Z}/4\mathbb{Z}$ . This contradiction proves the theorem.  $\square$

## References

- [1] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften 322, Springer, Berlin, 1999.

Table 1: The polynomials  $H_i$ , the constants  $c_i$  and, for the linear vertices,  $\eta = r_i(\xi)$ . Here  $\langle A, B \rangle = A + Bi$ .

| $i$ | $H_i$                                            | $c_i$                      | chart                                                |
|-----|--------------------------------------------------|----------------------------|------------------------------------------------------|
| 0   | $\langle x, 0 \rangle$                           | $\langle 54, 54 \rangle$   | $x = \langle 0, 0 \rangle$                           |
| 1   | $\langle -y - 1, 0 \rangle$                      | $\langle 1, 0 \rangle$     | $y = \langle -1, 0 \rangle$                          |
| 2   | $\langle x + 1, 0 \rangle$                       | $\langle 27, 0 \rangle$    | $x = \langle -1, 0 \rangle$                          |
| 3   | $\langle x, 1 \rangle$                           | $\langle -3, 3 \rangle$    | $x = \langle 0, -1 \rangle$                          |
| 4   | $\langle x, -1 \rangle$                          | $\langle 1, 1 \rangle$     | $x = \langle 0, 1 \rangle$                           |
| 5   | $\langle x - 1, 0 \rangle$                       | $\langle -27, -27 \rangle$ | $x = \langle 1, 0 \rangle$                           |
| 6   | $\langle -y + 1, 1 \rangle$                      | $\langle 1, 2 \rangle$     | $y = \langle 1, 1 \rangle$                           |
| 7   | $\langle -y + 1, -1 \rangle$                     | $\langle 1, 0 \rangle$     | $y = \langle 1, -1 \rangle$                          |
| 8   | $\langle -2y - 1, 1 \rangle$                     | $\langle -54, -78 \rangle$ | $2y = \langle -1, 1 \rangle$                         |
| 9   | $\langle -2y - 1, -1 \rangle$                    | $\langle -99, 18 \rangle$  | $2y = \langle -1, -1 \rangle$                        |
| 10  | $\langle x - 1, 1 \rangle$                       | $\langle -2, -2 \rangle$   | $x = \langle 1, -1 \rangle$                          |
| 11  | $\langle x - 1, -1 \rangle$                      | $\langle 0, 2 \rangle$     | $x = \langle 1, 1 \rangle$                           |
| 12  | $\langle x + y - 1, 0 \rangle$                   | $\langle 0, -3 \rangle$    | $y = \langle -x + 1, 0 \rangle$                      |
| 13  | $\langle 2x - y + 2, y \rangle$                  | $\langle -9, 0 \rangle$    | $\langle 1, 1 \rangle y = \langle 0, 2x + 2 \rangle$ |
| 14  | $\langle -y, x^2 \rangle$                        | $\langle 2, 0 \rangle$     | $y = \langle 0, x^2 \rangle$                         |
| 15  | $\langle -y, -x^2 \rangle$                       | $\langle 1, -1 \rangle$    | $y = \langle 0, -x^2 \rangle$                        |
| 16  | $\langle x^2 - 2y, 0 \rangle$                    | $\langle -2, -2 \rangle$   | $2y = \langle x^2, 0 \rangle$                        |
| 17  | $\langle -y + 1, -x^2 \rangle$                   | $\langle 0, 6 \rangle$     | $y = \langle 1, -x^2 \rangle$                        |
| 18  | $\langle x^2 - y, 1 \rangle$                     | $\langle 1, 1 \rangle$     | $y = \langle x^2, 1 \rangle$                         |
| 19  | $\langle -y, x^2 - 1 \rangle$                    | $\langle 0, 2 \rangle$     | $y = \langle 0, x^2 - 1 \rangle$                     |
| 20  | $\langle x^2 - y - 1, 0 \rangle$                 | $\langle 1, 0 \rangle$     | $y = \langle x^2 - 1, 0 \rangle$                     |
| 21  | $\langle -x^2 - y - 1, 0 \rangle$                | $\langle -1, 0 \rangle$    | $y = \langle -x^2 - 1, 0 \rangle$                    |
| 22  | $\langle x^2 - 2y, -x^2 \rangle$                 | $\langle 0, 2 \rangle$     | $2y = \langle x^2, -x^2 \rangle$                     |
| 23  | $\langle x^2 + x - y + 1, 0 \rangle$             | $\langle -6, 6 \rangle$    | $y = \langle x^2 + x + 1, 0 \rangle$                 |
| 24  | $\langle -x^2 - y + 1, x \rangle$                | $\langle -1, 0 \rangle$    | $y = \langle -x^2 + 1, x \rangle$                    |
| 25  | $\langle -x^2 - y + 1, -x \rangle$               | $\langle 1, 0 \rangle$     | $y = \langle -x^2 + 1, -x \rangle$                   |
| 26  | $\langle x^2 - x - y + 1, 0 \rangle$             | $\langle 1, 0 \rangle$     | $y = \langle x^2 - x + 1, 0 \rangle$                 |
| 27  | $\langle x - y, x^2 + 1 \rangle$                 | $\langle 0, 1 \rangle$     | $y = \langle x, x^2 + 1 \rangle$                     |
| 28  | $\langle -x^2 + x - y, 1 \rangle$                | $\langle 0, -1 \rangle$    | $y = \langle -x^2 + x, 1 \rangle$                    |
| 29  | $\langle -y, -x^2 - x + 1 \rangle$               | $\langle 2, -2 \rangle$    | $y = \langle 0, -x^2 - x + 1 \rangle$                |
| 30  | $\langle x - y, -x^2 - 1 \rangle$                | $\langle 1, 0 \rangle$     | $y = \langle x, -x^2 - 1 \rangle$                    |
| 31  | $\langle -y, x^2 + x - 1 \rangle$                | $\langle 1, 0 \rangle$     | $y = \langle 0, x^2 + x - 1 \rangle$                 |
| 32  | $\langle -x - y, x^2 - 1 \rangle$                | $\langle 0, 2 \rangle$     | $y = \langle -x, x^2 - 1 \rangle$                    |
| 33  | $\langle -x^2 - x - y, -1 \rangle$               | $\langle 0, 2 \rangle$     | $y = \langle -x^2 - x, -1 \rangle$                   |
| 34  | $\langle x - y - 1, x^2 \rangle$                 | $\langle 0, 1 \rangle$     | $y = \langle x - 1, x^2 \rangle$                     |
| 35  | $\langle x - y - 1, -x^2 \rangle$                | $\langle 1, 1 \rangle$     | $y = \langle x - 1, -x^2 \rangle$                    |
| 36  | $\langle -x^2 + x - y - 1, 0 \rangle$            | $\langle 0, 1 \rangle$     | $y = \langle -x^2 + x - 1, 0 \rangle$                |
| 37  | $\langle -y - 1, -x^2 - x \rangle$               | $\langle -2, 2 \rangle$    | $y = \langle -1, -x^2 - x \rangle$                   |
| 38  | $\langle -x^2 - x - y - 1, 0 \rangle$            | $\langle -2, 2 \rangle$    | $y = \langle -x^2 - x - 1, 0 \rangle$                |
| 39  | $\langle x^2 - x - 2y - 1, 0 \rangle$            | $\langle 6, 6 \rangle$     | $2y = \langle x^2 - x - 1, 0 \rangle$                |
| 40  | $\langle x^2 + x - 2y - 1, x^2 + x + 1 \rangle$  | $\langle 1, 1 \rangle$     | $2y = \langle x^2 + x - 1, x^2 + x + 1 \rangle$      |
| 41  | $\langle x^2 + x - 2y - 1, -x^2 + x + 1 \rangle$ | $\langle 1, 1 \rangle$     | $2y = \langle x^2 + x - 1, -x^2 + x + 1 \rangle$     |
| 42  | $\langle -x^2 - x - 2y - 1, x^2 + x + 1 \rangle$ | $\langle 0, 1 \rangle$     | $2y = \langle -x^2 - x - 1, x^2 + x + 1 \rangle$     |
| 43  | $\langle x^2 - x - 2y - 1, x^2 + x - 1 \rangle$  | $\langle 1, 0 \rangle$     | $2y = \langle x^2 - x - 1, x^2 + x - 1 \rangle$      |
| 44  | $\langle -y, 0 \rangle$                          | $\langle 1, 0 \rangle$     | $y = \langle 0, 0 \rangle$                           |
| 45  | $\langle -y - 2, 0 \rangle$                      | $\langle -54, -27 \rangle$ | $y = \langle -2, 0 \rangle$                          |
| 46  | $\langle x - y, -x + 1 \rangle$                  | $\langle -3, -4 \rangle$   | $y = \langle x, -x + 1 \rangle$                      |
| 47  | $\langle x - y, 1 \rangle$                       | $\langle 2, -2 \rangle$    | $y = \langle x, 1 \rangle$                           |
| 48  | $\langle x - 2, 0 \rangle$                       | $\langle 3, 0 \rangle$     | $x = \langle 2, 0 \rangle$                           |
| 49  | $\langle x + 2, 0 \rangle$                       | $\langle -6, 6 \rangle$    | $x = \langle -2, 0 \rangle$                          |
| 50  | $\langle x - y, -x \rangle$                      | $\langle 22, -4 \rangle$   | $y = \langle x, -x \rangle$                          |
| 51  | $\langle -y + 1, x^2 \rangle$                    | $\langle 6, -6 \rangle$    | $y = \langle 1, x^2 \rangle$                         |
| 52  | $\langle x^2 + x - y, 1 \rangle$                 | $\langle -2, 0 \rangle$    | $y = \langle x^2 + x, 1 \rangle$                     |
| 53  | $\langle x, -2 \rangle$                          | $\langle 18, 18 \rangle$   | $x = \langle 0, 2 \rangle$                           |



| $i$ | $H_i$                                  | $c_i$                     | chart                                                 |
|-----|----------------------------------------|---------------------------|-------------------------------------------------------|
| 54  | $\langle x - y + 1, x^2 \rangle$       | $\langle -54, 54 \rangle$ | $y = \langle x + 1, x^2 \rangle$                      |
| 55  | $\langle -x^2 + x - y - 1, -2 \rangle$ | $\langle -1, -1 \rangle$  | $y = \langle -x^2 + x - 1, -2 \rangle$                |
| 56  | $\langle x - y - 1, x^2 - x \rangle$   | $\langle 1, 0 \rangle$    | $y = \langle x - 1, x^2 - x \rangle$                  |
| 57  | $\langle x - y + 1, x^2 + 2 \rangle$   | $\langle 3, -3 \rangle$   | $y = \langle x + 1, x^2 + 2 \rangle$                  |
| 58  | $\langle -y - 1, -x \rangle$           | $\langle 0, 2 \rangle$    | $y = \langle -1, -x \rangle$                          |
| 59  | $\langle x, y + 1 \rangle$             | $\langle 0, 2 \rangle$    | $y = \langle -1, x \rangle$                           |
| 60  | $\langle -y + 1, 0 \rangle$            | $\langle 3, -3 \rangle$   | $y = \langle 1, 0 \rangle$                            |
| 61  | $\langle x + 1, -y \rangle$            | $\langle 1, -3 \rangle$   | $y = \langle 0, -x - 1 \rangle$                       |
| 62  | $\langle 2x - y - 1, y - 1 \rangle$    | $\langle 2, -11 \rangle$  | $\langle 1, 1 \rangle y = \langle 1, 2x - 1 \rangle$  |
| 63  | $\langle -y, x + 1 \rangle$            | $\langle 14, 2 \rangle$   | $y = \langle 0, x + 1 \rangle$                        |
| 64  | $\langle x, 2 \rangle$                 | $\langle -54, 54 \rangle$ | $x = \langle 0, -2 \rangle$                           |
| 65  | $\langle -x - y, x + 1 \rangle$        | $\langle 0, 2 \rangle$    | $y = \langle -x, x + 1 \rangle$                       |
| 66  | $\langle 2x + y + 1, -y + 1 \rangle$   | $\langle -2, 2 \rangle$   | $\langle 1, 1 \rangle y = \langle 1, -2x - 1 \rangle$ |
| 67  | $\langle -y + 1, -2 \rangle$           | $\langle 3, 0 \rangle$    | $y = \langle 1, -2 \rangle$                           |
| 68  | $\langle -x - y, 1 \rangle$            | $\langle -2, 2 \rangle$   | $y = \langle -x, 1 \rangle$                           |
| 69  | $\langle -y - 1, 2 \rangle$            | $\langle 0, 2 \rangle$    | $y = \langle -1, 2 \rangle$                           |
| 70  | $\langle x - y - 1, 2 \rangle$         | $\langle 1, 0 \rangle$    | $y = \langle x - 1, 2 \rangle$                        |
| 71  | $\langle -y, 2 \rangle$                | $\langle 1, 0 \rangle$    | $y = \langle 0, 2 \rangle$                            |
| 72  | $\langle x - y - 1, -2 \rangle$        | $\langle 0, -1 \rangle$   | $y = \langle x - 1, -2 \rangle$                       |
| 73  | $\langle -y, -2 \rangle$               | $\langle -1, -1 \rangle$  | $y = \langle 0, -2 \rangle$                           |
| 74  | $\langle -x - y - 1, 1 \rangle$        | $\langle -2, 2 \rangle$   | $y = \langle -x - 1, 1 \rangle$                       |
| 75  | $\langle -y + 1, x + 1 \rangle$        | $\langle 0, -1 \rangle$   | $y = \langle 1, x + 1 \rangle$                        |
| 76  | $\langle -x - y, -x + 1 \rangle$       | $\langle 1, 0 \rangle$    | $y = \langle -x, -x + 1 \rangle$                      |
| 77  | $\langle x - y, 0 \rangle$             | $\langle 1, 0 \rangle$    | $y = \langle x, 0 \rangle$                            |
| 78  | $\langle x, y \rangle$                 | $\langle 6, -2 \rangle$   | $y = \langle 0, x \rangle$                            |
| 79  | $\langle -y + 1, -x \rangle$           | $\langle 0, 1 \rangle$    | $y = \langle 1, -x \rangle$                           |
| 80  | $\langle 2x - y, y \rangle$            | $\langle 2, 0 \rangle$    | $\langle 1, 1 \rangle y = \langle 0, 2x \rangle$      |
| 81  | $\langle -y + 1, 2 \rangle$            | $\langle 0, 18 \rangle$   | $y = \langle 1, 2 \rangle$                            |

Table 2: The data of the identities (2); the entry  $j:e$  means  $\rho_{ij} = e \neq 0$ .

|                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $i = 0, D_i = 1, \rho_{ij}: 9:2, 13:2, 16:2, 23:1, 31:3, 39:1, 41:1, 55:1, 67:1, 69:1, 73:2, 79:2$                                                                        |
| $w_i \quad \langle -18t^2 - 138t - 24, 78t^2 - 30t - 12 \rangle$                                                                                                          |
| $i = 1, D_i = 1, \rho_{ij}: 14:1, 18:3, 22:2, 25:2, 26:1, 29:2, 31:3, 35:1, 37:2, 41:3, 48:3, 55:3, 61:1, 62:1, 66:3, 75:1, 77:3, 80:2$                                   |
| $w_i \quad \langle 544t^4 + 296t^3 - 224t^2 + 616t + 1696, 336t^4 + 600t^3 + 192t^2 - 696t - 608 \rangle$                                                                 |
| $i = 2, D_i = 1, \rho_{ij}: 8:2, 12:2, 17:1, 18:3, 20:1, 21:3, 25:3, 26:3, 33:2, 34:2, 35:2, 38:2, 39:1, 45:3, 52:1, 62:2, 69:1, 71:1, 73:2, 75:3$                        |
| $w_i \quad \langle 1035t^2 + 3096t + 837, 1035t^2 + 3096t + 837 \rangle$                                                                                                  |
| $i = 3, D_i = 1, \rho_{ij}: 7:3, 9:1, 16:1, 25:1, 32:3, 34:1, 38:3, 40:1, 41:2, 54:1, 57:1, 58:3, 65:3, 68:2, 73:1, 74:2, 80:3$                                           |
| $w_i \quad \langle -30t^2 - 300t - 30, -72t^2 - 102t - 6 \rangle$                                                                                                         |
| $i = 4, D_i = 1, \rho_{ij}: 6:3, 7:2, 16:3, 20:3, 25:1, 28:2, 29:3, 31:1, 35:2, 38:2, 42:3, 44:3, 57:3, 59:3, 67:2, 73:2, 75:2, 81:2$                                     |
| $w_i \quad \langle 936t^2 - 2259t - 567, -828t^2 - 1161t - 225 \rangle$                                                                                                   |
| $i = 5, D_i = 1, \rho_{ij}: 9:2, 12:1, 13:2, 14:3, 16:3, 17:3, 25:3, 26:1, 27:2, 35:3, 36:2, 39:2, 63:3, 66:3, 67:3, 72:3$                                                |
| $w_i \quad \langle 666t^2 + 1692t + 450, -864t^2 + 1512t + 360 \rangle$                                                                                                   |
| $i = 6, D_i = 1, \rho_{ij}: 4:1, 10:1, 11:2, 14:1, 20:3, 25:1, 28:3, 30:1, 32:1, 33:2, 35:2, 40:3, 41:2, 42:3, 48:1, 51:2, 52:3, 53:2, 56:2, 64:3, 66:2, 80:1$            |
| $w_i \quad \langle 1766t^4 - 4614t^3 + 528t^2 + 7602t - 12326, 162t^4 + 2502t^3 + 96t^2 + 2814t - 2082 \rangle$                                                           |
| $i = 7, D_i = 1, \rho_{ij}: 3:1, 4:2, 11:3, 28:3, 29:1, 32:2, 35:3, 38:1, 39:2, 40:3, 41:2, 42:2, 43:1, 48:1, 50:2, 53:3, 55:3, 57:1, 62:2, 63:3, 65:2, 75:1, 77:3, 78:3$ |
| $w_i \quad \langle 6954t^4 - 102t^3 - 1218t^2 - 21288t + 15084, 2622t^4 + 4314t^3 + 18426t^2 - 7584t - 288 \rangle$                                                       |
| $i = 8, D_i = 8, e_i = 2, \rho_{ij}: 2:2, 11:3, 15:2, 32:3, 40:1, 41:1, 42:2, 48:2, 49:3, 50:3, 57:3, 59:2, 62:3, 70:2, 79:2, 80:2$                                       |
| $w_i \quad \langle -82176t^4 + 148992t^3 + 148992t^2 + 181632t + 224640, -76032t^4 + 44544t^3 + 140544t^2 - 206976t - 155520 \rangle$                                     |
| $i = 9, D_i = 16, e_i = 3, \rho_{ij}: 0:2, 3:3, 5:2, 10:1, 16:3, 18:2, 21:3, 32:1, 35:1, 37:3, 38:2, 41:2, 46:3, 48:2, 49:2, 52:2, 58:1, 64:2, 72:2$                      |

---


$$\begin{aligned}
w_i & \langle -13824t^4 - 61824t^3 + 764928t^2 + 1617024t - 324864, \\
& \quad 19968t^4 - 220032t^3 - 880896t^2 + 853632t + 1045248 \rangle \\
i = 10, D_i = 1, \rho_{ij}: & 6:3, 9:3, 14:2, 15:1, 20:2, 30:2, 36:1, 41:2, 45:2, 51:3, 61:1, 62:1, 69:3, 70:1, 72:2, 77:3, 78:2, 80:3 \\
w_i & \langle -4306t^2 + 9610t + 4984, \\
& \quad -3486t^2 - 9358t + 12528 \rangle \\
i = 11, D_i = 1, \rho_{ij}: & 6:2, 7:1, 8:1, 14:3, 15:1, 41:1, 45:1, 54:3, 60:3, 69:2, 70:1, 76:1 \\
w_i & \langle 39t^2 + 28t - 105, \\
& \quad 11t^2 - 104t + 23 \rangle \\
i = 12, D_i = 1, \rho_{ij}: & 2:2, 5:3, 16:2, 18:1, 25:3, 31:1, 34:2, 35:2, 38:3, 41:2, 48:3, 49:3, 61:1, 63:1, 64:1, 66:2, 72:1, 77:1 \\
w_i & \langle 218t^4 - 372t^3 + 980t^2 - 1632t + 912, \\
& \quad -50t^4 - 220t^3 + 298t^2 + 236t - 300 \rangle \\
i = 13, D_i = 4, e_i = 4, \rho_{ij}: & 0:2, 5:2, 19:2, 25:2, 41:2, 63:2, 69:2, 71:2 \\
w_i & \langle 16t^4 - 96t^3 - 96t^2 - 144t - 192, \\
& \quad -112t^4 - 96t^3 - 160t^2 - 144t + 96 \rangle \\
i = 14, D_i = 1, \rho_{ij}: & 1:3, 5:1, 6:3, 10:2, 11:1, 26:1, 35:3, 41:1, 45:1, 46:3, 52:3, 60:1, 64:1, 66:1, 69:3 \\
w_i & \langle 22t^5 + 93t^4 + 94t^3 - 226t^2 - 142t + 143, \\
& \quad 8t^5 - 287t^4 + 60t^3 + 578t^2 - 46t + 39 \rangle \\
i = 15, D_i = 1, \rho_{ij}: & 8:2, 10:3, 11:3, 24:1, 29:2, 30:2, 34:1, 45:3, 46:3, 53:2, 60:3, 62:1, 63:1, 65:2, 69:1, 73:3 \\
w_i & \langle -15t^5 + 25t^4 - 139t^3 + 51t^2 + 62t - 56, \\
& \quad -t^5 - 83t^4 - 71t^3 + 243t^2 - 42t + 10 \rangle \\
i = 16, D_i = 512, e_i = 1, \rho_{ij}: & 0:2, 3:3, 4:1, 5:1, 9:1, 12:2, 20:3, 21:1, 23:3, 25:1, 36:3, 37:3, 49:3, 58:1, 59:2, 64:1, 65:3, 68:1, \\
& 69:3, 78:1, 79:3 \\
w_i & \langle -2501512472461312t^5 - 84767359762432t^4 + 675266415296512t^3 - 5379248952049664t^2 \\
& \quad + 2827508431585280t - 2512153466896384, \\
& \quad -1463864653479936t^5 - 49605161582592t^4 + 395160389025792t^3 - 3147892521435136t^2 \\
& \quad + 1654634827874304t - 1470091677663232 \rangle \\
i = 17, D_i = 1, \rho_{ij}: & 2:3, 5:1, 18:3, 29:2, 30:1, 45:1, 49:1, 50:2, 51:1, 53:3, 56:1, 58:1, 59:1, 64:1, 67:3, 81:2 \\
w_i & \langle 42t^5 - 6t^4 - 84t^3 + 54t^2 + 24t + 24, \\
& \quad 24t^5 - 36t^4 + 18t^3 + 24t^2 - 96t + 48 \rangle \\
i = 18, D_i = 1, \rho_{ij}: & 1:1, 2:1, 9:2, 12:3, 17:1, 34:1, 35:3, 46:1, 47:2, 53:1, 58:3, 64:3, 65:2, 71:2, 75:3, 76:1, 77:3 \\
w_i & \langle -3058t^5 + 1151t^4 + 15t^3 - 1072t^2 + 496t - 532, \\
& \quad 1198t^5 - 4617t^4 + 683t^3 - 1178t^2 + 2160t - 4636 \rangle \\
i = 19, D_i = 1, \rho_{ij}: & 13:2, 20:2, 25:2, 42:2, 58:2, 70:2, 73:2 \\
w_i & \langle 2t^5 + 2t^4 + 2t^3 + 2t^2 - 2t, \\
& \quad 2t^5 - 8t^3 + 6t \rangle \\
i = 20, D_i = 1, \rho_{ij}: & 2:3, 4:1, 6:1, 10:2, 16:1, 19:2, 22:2, 24:1, 25:3, 35:3, 36:1, 38:1, 39:3, 43:3, 50:2, 51:1, 56:1, 59:1, 63:2, 65:3, \\
& 66:3, 68:1 \\
w_i & \langle -128t^5 - 26t^4 + 1838t^3 + 2436t^2 - 1544t - 1888, \\
& \quad -624t^5 - 1158t^4 + 1394t^3 + 3088t^2 - 832t - 1664 \rangle \\
i = 21, D_i = 1, \rho_{ij}: & 2:1, 9:1, 16:3, 23:3, 26:1, 28:3, 32:3, 39:3, 42:1, 45:2, 47:1, 53:2, 58:3, 60:1, 64:1, 65:2, 66:1, 69:2, 77:1, 80:1 \\
w_i & \langle -16257t^5 - 14353t^4 - 30895t^3 - 2285t^2 - 12293t + 7308, \\
& \quad 2987t^5 - 8965t^4 - 8937t^3 - 27087t^2 - 9965t - 14708 \rangle \\
i = 22, D_i = 4, e_i = 0, \rho_{ij}: & 1:2, 20:2, 30:2, 35:2, 39:2, 68:2, 79:2, 80:2 \\
w_i & \langle 288t^5 + 64t^4 + 64t^3 + 512t^2 - 192t + 320, \\
& \quad 32t^5 + 192t^4 - 256t^3 - 576t^2 - 64t - 64 \rangle \\
i = 23, D_i = 7776, \rho_{ij}: & 0:3, 16:1, 21:1, 29:1, 30:2, 37:1, 39:2, 44:2, 45:1, 49:2, 57:3, 59:1, 60:2, 66:3 \\
w_i & \langle -186624t^5 - 746496t^4 - 1119744t^3 - 2052864t^2 - 1306368t - 1119744, \\
& \quad 435456t^5 + 186624t^4 + 1181952t^2 + 746496t + 1492992 \rangle \\
i = 24, D_i = 100000000, \rho_{ij}: & 15:3, 20:3, 26:2, 34:3, 36:1, 37:2, 39:1, 44:3, 55:1, 59:2, 62:3, 68:1, 70:1, 73:1, 76:3, 77:3 \\
w_i & \langle 3877200000000t^5 - 4624000000000t^4 - 1189800000000t^3 + 8568800000000t^2 + 15513600000000t \\
& \quad - 3180000000000, \\
& \quad -6797200000000t^5 - 1191280000000t^4 + 8406800000000t^3 + 23472800000000t^2 \\
& \quad - 4975200000000t - 17253600000000 \rangle \\
i = 25, D_i = 500000000000, \rho_{ij}: & 1:2, 2:1, 3:3, 4:3, 5:1, 6:3, 12:1, 13:2, 16:3, 19:2, 20:1, 31:1, 32:3, 35:3, 39:3, 45:3, 47:3, 52:2, \\
& 54:2, 61:3, 64:1, 69:3, 71:2, 77:1, 78:1, 80:2 \\
w_i & \langle -437810920000000000000t^5 - 900468536000000000000t^4 + 461285040000000000000t^3 \\
& \quad + 1768252664000000000000t^2 - 181584200000000000000t - 1260908352000000000000, \\
& \quad 3516292400000000000000t^5 - 1921652720000000000000t^4 - 944103648000000000000t^3 \\
& \quad + 3449767600000000000000t^2 + 1161409400000000000000t - 345413280000000000000 \rangle \\
i = 26, D_i = 1, \rho_{ij}: & 1:3, 2:1, 5:3, 14:3, 21:3, 24:2, 27:3, 28:3, 29:3, 31:3, 34:2, 35:3, 36:2, 41:3, 43:3, 44:1, 46:3, 54:3, 55:3, 58:1, \\
& 59:3, 62:3, 64:2, 65:2, 66:1, 67:1, 68:3, 69:2, 71:1, 73:3, 79:3 \\
w_i & \langle 12107776t^5 - 2885632t^4 + 7415296t^3 + 3623424t^2 + 684032t + 15578112, \\
& \quad 1517056t^5 - 14114304t^4 + 11950592t^3 - 20907520t^2 + 21449728t - 23347200 \rangle \\
i = 27, D_i = 128, \rho_{ij}: & 5:2, 26:1, 29:2, 31:2, 38:1, 43:3, 45:2, 46:1, 50:2, 55:2, 56:3, 59:1, 62:2, 69:1, 71:1, 75:3, 76:1, 79:1, 80:3 \\
w_i & \langle 16896t^5 + 71168t^4 + 52224t^3 - 22016t^2 - 48640t - 51200, \\
& \quad 20992t^5 - 24064t^4 - 105472t^3 - 56832t^2 - 33280t - 40960 \rangle \\
i = 28, D_i = 7776, \rho_{ij}: & 4:2, 6:1, 7:1, 21:1, 26:1, 31:2, 32:1, 46:1, 50:1, 51:1, 54:3, 62:1, 64:3, 67:3, 71:2, 72:1 \\
w_i & \langle -8118144t^5 + 14852160t^4 - 20684160t^3 + 14338944t^2 + 1477440t + 7402752, \\
& \quad -8118144t^5 + 14852160t^4 - 20684160t^3 + 14338944t^2 + 1477440t + 7402752 \rangle
\end{aligned}$$


---

---


$$\begin{aligned}
& -13250304t^5 + 6111936t^4 + 12752640t^3 - 7620480t^2 - 4525632t - 13405824) \\
i = 29, D_i = 2560, \rho_{ij}: & 1:2, 4:1, 7:3, 15:2, 17:2, 23:3, 26:1, 27:2, 32:2, 41:1, 45:2, 47:3, 50:2, 55:1, 58:3, 61:1, 65:1, 71:2, 72:3 \\
w_i & \langle 84736000t^5 + 165847040t^4 - 304824320t^3 - 283822080t^2 - 82595840t + 150159360, \\
& -436070400t^5 - 109547520t^4 + 517611520t^3 + 389048320t^2 + 315985920t - 442552320 \rangle \\
i = 30, D_i = 256, \rho_{ij}: & 6:3, 10:2, 15:2, 17:3, 22:2, 23:2, 31:2, 33:2, 41:2, 46:1, 47:2, 50:3, 52:1, 62:3, 63:2, 73:2, 78:1, 80:3 \\
w_i & \langle 59392t^5 + 38912t^4 + 24576t^3 + 337920t^2 + 112640t + 225280, \\
& -200704t^5 - 88064t^4 - 253952t^3 + 81920t^2 + 51200t + 40960 \rangle \\
i = 31, D_i = 1280, \rho_{ij}: & 0:1, 1:1, 4:3, 12:3, 25:3, 26:1, 27:2, 28:2, 30:2, 37:2, 41:1, 42:2, 53:1, 63:2, 66:3, 70:2, 75:3, 78:3 \\
w_i & \langle -209377280t^5 + 90572800t^4 + 48711680t^3 - 20520960t^2 + 131706880t - 149002240, \\
& 482437120t^5 + 218296320t^4 - 708761600t^3 - 570112000t^2 - 365752320t + 533841920 \rangle \\
i = 32, D_i = 839808, \rho_{ij}: & 3:1, 6:3, 7:2, 8:1, 9:3, 21:1, 25:1, 28:3, 29:2, 38:2, 44:2, 49:3, 50:1, 62:2, 65:1, 68:2, 73:3, 75:1, 78:1, \\
& 80:3 \\
w_i & \langle -180948390912t^5 + 196347110400t^4 + 423807427584t^3 - 114731209728t^2 + 79593643008t \\
& -232774622208, \\
& -141188520960t^5 - 255852546048t^4 + 334411545600t^3 + 170507897856t^2 + 19658225664t \\
& +155807898624 \rangle \\
i = 33, D_i = 4, \rho_{ij}: & 2:2, 6:2, 30:2, 36:2, 53:2, 64:2, 66:2, 69:2 \\
w_i & \langle -8t^5 + 24t^2 + 8t + 24, \\
& -8t^5 - 40t^2 - 40t + 8 \rangle \\
i = 34, D_i = 256, \rho_{ij}: & 2:2, 3:3, 12:2, 15:3, 18:3, 24:1, 26:2, 38:1, 42:1, 44:1, 48:1, 51:2, 53:2, 55:1, 56:2, 60:1, 62:1, 64:1, 77:2, \\
& 78:3, 81:2 \\
w_i & \langle -1374208t^5 + 1791488t^4 + 181760t^3 - 1800192t^2 + 5194752t - 2566144, \\
& -912384t^5 + 3259904t^4 - 6157824t^3 + 5612544t^2 - 2456576t + 1562624 \rangle \\
i = 35, D_i = 512, \rho_{ij}: & 1:3, 2:2, 4:2, 5:1, 6:2, 7:1, 9:3, 12:2, 14:1, 18:1, 20:1, 22:2, 25:1, 26:1, 37:2, 44:2, 45:3, 46:1, 48:1, 53:2, \\
& 60:3, 62:2, 64:2, 70:3, 77:3, 79:2 \\
w_i & \langle -235714560t^5 - 100411392t^4 + 1278009344t^3 - 1663442944t^2 + 2056011776t - 1082146816, \\
& 549316608t^5 - 1338644480t^4 + 1831215104t^3 - 1338449920t^2 - 292298752t + 40370176 \rangle \\
i = 36, D_i = 60466176, \rho_{ij}: & 5:2, 10:3, 16:1, 20:3, 24:3, 26:2, 33:2, 37:1, 45:1, 46:3, 48:3, 49:2, 53:3, 57:3, 59:2, 61:1, 62:3, 64:3, \\
& 65:1, 66:1, 68:1, 69:1, 73:1, 79:2 \\
w_i & \langle -2913018494976t^5 + 9168123469824t^4 - 4806819127296t^3 + 15926307028992t^2 - 7392352813056t \\
& +8267903041536, \\
& -1220933025792t^5 + 1541645623296t^4 - 1660643057664t^3 + 3397715361792t^2 - 2094548336640t \\
& +1751100456960 \rangle \\
i = 37, D_i = 500000000, \rho_{ij}: & 1:2, 9:1, 16:1, 23:3, 24:2, 31:2, 35:2, 36:3, 42:1, 45:1, 47:2, 49:3, 50:2, 53:2, 62:3, 64:2, 66:3, 76:3, \\
& 78:2 \\
w_i & \langle -1432000000000t^5 - 188000000000t^4 - 2004000000000t^3 + 608000000000t^2 + 1176000000000t \\
& -276800000000, \\
& 2360000000000t^5 - 588000000000t^4 + 693200000000t^3 + 142400000000t^2 - 314400000000t \\
& -632000000000 \rangle \\
i = 38, D_i = 4096, \rho_{ij}: & 2:2, 3:1, 4:2, 7:3, 9:2, 12:1, 20:3, 27:3, 32:2, 34:3, 44:3, 48:1, 49:1, 51:2, 54:2, 56:3, 60:2, 63:1, 64:2, \\
& 65:2, 66:2, 67:1, 74:3, 75:3, 79:1, 80:1, 81:2 \\
w_i & \langle -70189056t^5 - 147881984t^4 - 86016000t^3 - 311230464t^2 - 377880576t - 219283456, \\
& -47513600t^5 + 94076928t^4 - 60719104t^3 - 101515264t^2 - 57540608t + 52822016 \rangle \\
i = 39, D_i = 128, e_i = 2, \rho_{ij}: & 0:3, 2:3, 5:2, 7:2, 20:1, 21:1, 22:2, 23:2, 24:3, 25:1, 44:2, 49:2, 50:1, 52:2, 53:3, 64:3, 79:2, 80:3 \\
w_i & \langle -16900169728t^5 + 556081152t^4 - 2778832896t^3 - 70577070080t^2 + 84685234176t + 30231060480, \\
& 5867675648t^5 - 189521920t^4 + 973414400t^3 + 24501018624t^2 - 29413711872t - 10498940928 \rangle \\
i = 40, D_i = 4, e_i = 3, \rho_{ij}: & 3:3, 6:1, 7:1, 8:3, 44:1, 62:3 \\
w_i & \langle -32t^5 - 32t^4 - 64t^3 - 32t^2 + 32t, \\
& 32t^5 - 32t^4 - 64t^3 - 96t^2 - 96t \rangle \\
i = 41, D_i = 25000000000, e_i = 1, \rho_{ij}: & 0:3, 1:1, 3:2, 6:2, 7:2, 8:3, 9:2, 10:2, 11:3, 12:2, 13:2, 14:3, 26:1, 29:3, 30:2, 31:3, 45:3, \\
& 59:1, 60:1, 61:3, 66:2, 70:3, 77:3, 80:1 \\
w_i & \langle -9593203739238400000000000t^5 + 8128809416806400000000000t^4 \\
& +170429135882240000000000000t^3 - 281848236300288000000000000t^2 \\
& +67446062611456000000000000t + 22165952113459200000000000, \\
& -1222921823283200000000000000t^5 - 151440848365568000000000000t^4 \\
& +39551938741248000000000000t^3 + 88656606772224000000000000t^2 \\
& -160779465424896000000000000t - 2401180920729600000000000 \rangle \\
i = 42, D_i = 20000000, e_i = 3, \rho_{ij}: & 4:1, 6:1, 7:2, 8:2, 19:2, 21:3, 31:2, 34:3, 37:3, 45:1, 50:3, 56:2, 64:2, 70:1, 73:1, 77:3 \\
w_i & \langle -2133995520000000t^5 - 2664483840000000t^4 - 3559736320000000t^3 - 4616263680000000t^2 \\
& -1595632640000000t - 2414423040000000, \\
& 1872117760000000t^5 + 718504960000000t^4 + 1793745920000000t^3 - 2945771520000000t^2 \\
& -4889000960000000t - 3160826880000000 \rangle \\
i = 43, D_i = 16, e_i = 0, \rho_{ij}: & 7:3, 20:1, 26:1, 27:1, 52:3, 78:3 \\
w_i & \langle -128t^5 - 128t^4 + 128t^3 + 128t^2 + 128t - 128, \\
& 64t^5 - 128t^4 - 192t^3 + 128t^2 + 64t + 256 \rangle \\
i = 44, D_i = 1, \rho_{ij}: & 4:1, 23:2, 24:1, 26:3, 32:2, 34:3, 35:2, 38:1, 39:2, 40:3, 50:1, 55:3, 56:2, 59:3, 76:2 \\
w_i & \langle 29t^4 - 29t^3 + 37t^2 - 46t + 26, \\
& -11t^4 + 13t^3 - 23t^2 + 26t - 12 \rangle
\end{aligned}$$


---

---

$i = 45, D_i = 1, \rho_{ij}: 2:1, 10:2, 11:3, 14:3, 15:1, 17:3, 21:2, 23:3, 25:1, 27:2, 29:2, 35:1, 36:3, 37:3, 41:1, 42:3, 48:3, 50:1, 51:2, 62:1, 65:2, 66:2, 76:1, 77:1$   
 $w_i \quad \langle 10200t^4 + 24150t^3 - 61950t^2 - 33150t - 45000, \\ -40200t^4 - 20700t^3 - 6150t^2 + 202950t - 12150 \rangle$

$i = 46, D_i = 1, \rho_{ij}: 9:1, 14:1, 15:1, 18:3, 26:1, 27:3, 28:3, 30:3, 35:3, 36:1, 49:3, 54:1, 57:1, 64:2, 70:2, 78:1$   
 $w_i \quad \langle 995t^4 + 219t^3 + 1489t^2 - 479t - 214, \\ -895t^4 + 737t^3 + 927t^2 + 233t + 648 \rangle$

$i = 47, D_i = 1, \rho_{ij}: 18:2, 21:3, 25:1, 29:1, 30:2, 37:2, 63:1$   
 $w_i \quad \langle 3t^3 + 6t^2 + 6t + 6, \\ t^3 - 2t^2 - 2 \rangle$

$i = 48, D_i = 1, \rho_{ij}: 1:1, 6:3, 7:3, 8:2, 9:2, 12:1, 34:3, 35:3, 36:1, 38:3, 45:1, 50:2, 60:3, 65:2, 66:2, 70:1, 79:1, 80:2$   
 $w_i \quad \langle 25272t^2 - 6858t - 19494, \\ 0 \rangle$

$i = 49, D_i = 1, \rho_{ij}: 8:1, 9:2, 12:1, 16:1, 17:3, 23:2, 32:1, 36:2, 37:1, 38:3, 39:2, 46:1, 51:2, 58:2, 69:1, 74:1, 78:3$   
 $w_i \quad \langle -4230t^2 + 7272t + 13374, \\ 972t^2 + 1296t - 9756 \rangle$

$i = 50, D_i = 1, \rho_{ij}: 7:2, 8:1, 17:2, 20:2, 27:2, 28:3, 29:2, 30:1, 32:3, 37:2, 39:3, 42:1, 44:3, 45:3, 48:2, 55:1, 56:2, 66:1, 70:3, 73:3$   
 $w_i \quad \langle 63052t^4 + 150232t^3 + 90428t^2 + 142136t - 7092, \\ -52824t^4 - 69724t^3 + 61244t^2 + 169708t + 40504 \rangle$

$i = 51, D_i = 1, \rho_{ij}: 6:2, 10:1, 17:3, 20:3, 28:3, 34:2, 38:2, 45:2, 49:2, 58:2, 59:3, 64:2, 67:2, 74:3, 77:3, 79:3, 81:2$   
 $w_i \quad \langle -1220t^5 - 3168t^4 - 3786t^3 + 23082t^2 + 8408t + 1464, \\ -6380t^5 + 6744t^4 + 6306t^3 + 4842t^2 + 7448t - 25320 \rangle$

$i = 52, D_i = 216, \rho_{ij}: 2:3, 6:1, 9:2, 14:1, 25:2, 30:3, 39:2, 43:1, 71:1$   
 $w_i \quad \langle 864t^5 + 1728t^4 + 4752t^3 + 6912t^2 + 1728t + 2160, \\ -1728t^5 - 1728t^4 - 432t^3 + 864t^2 + 1728t + 2160 \rangle$

$i = 53, D_i = 1, \rho_{ij}: 6:2, 7:1, 15:2, 17:1, 18:3, 21:2, 31:3, 33:2, 34:2, 35:2, 36:1, 37:2, 39:1, 54:2, 60:1, 62:2, 65:3, 71:1, 78:1, 80:3$   
 $w_i \quad \langle 53694t^2 + 90738t + 306540, \\ -47214t^2 - 190260t - 781794 \rangle$

$i = 54, D_i = 32, \rho_{ij}: 3:3, 11:1, 25:2, 26:1, 28:1, 38:2, 46:3, 53:2, 56:3, 60:3, 63:1, 71:3$   
 $w_i \quad \langle 960t^5 + 5184t^4 + 10752t^3 + 9984t^2 + 1536t, \\ 1728t^5 + 3264t^4 - 768t^3 - 6912t^2 - 11520t - 6912 \rangle$

$i = 55, D_i = 3359232, \rho_{ij}: 0:3, 1:1, 7:1, 24:3, 26:1, 27:2, 29:3, 34:3, 44:1, 50:3, 56:3, 61:3, 73:2, 76:3, 77:2, 79:2$   
 $w_i \quad \langle 3125456326656t^5 + 4848124243968t^4 - 13881824686080t^3 + 30173454913536t^2 \\ - 22864652918784t + 13280038428672, \\ 5890023641088t^5 - 9420603346944t^4 + 18631563743232t^3 - 3001836589056t^2 - 4167651336192t \\ + 14772988624896 \rangle$

$i = 56, D_i = 31250, \rho_{ij}: 6:2, 17:3, 20:3, 27:1, 34:2, 38:1, 42:2, 44:2, 50:2, 54:1, 55:1, 60:2, 64:2, 77:3$   
 $w_i \quad \langle -30500000t^5 + 42812500t^4 + 22375000t^3 - 118437500t^2 + 99125000t - 12250000, \\ 13500000t^5 + 69312500t^4 - 105562500t^3 + 57812500t^2 + 30812500t - 47000000 \rangle$

$i = 57, D_i = 32, \rho_{ij}: 3:3, 4:1, 7:3, 8:1, 23:1, 36:1, 46:3, 59:3, 62:1, 63:1, 64:3, 70:2, 72:3, 81:2$   
 $w_i \quad \langle -17856t^5 + 5184t^4 - 74688t^3 - 24960t^2 - 7296t - 36864, \\ 2688t^5 + 6912t^4 + 82368t^3 + 19200t^2 + 120192t + 53760 \rangle$

$i = 58, D_i = 1, \rho_{ij}: 3:1, 9:3, 16:3, 17:3, 18:1, 19:2, 21:1, 26:3, 29:1, 49:2, 51:2, 65:1, 67:1, 68:3, 74:3$   
 $w_i \quad \langle 26t^4 - 284t^3 - 660t^2 - 276t + 456, \\ -156t^4 - 110t^3 + 160t^2 + 1088t + 264 \rangle$

$i = 59, D_i = 1, \rho_{ij}: 4:1, 8:2, 16:2, 17:3, 20:3, 23:3, 24:2, 26:1, 27:3, 36:2, 41:3, 44:1, 51:1, 57:1, 66:1, 77:2$   
 $w_i \quad \langle 88t^4 + 174t^3 + 1060t^2 - 1982t - 1240, \\ -368t^4 - 394t^3 + 782t^2 + 2074t - 864 \rangle$

$i = 60, D_i = 1, \rho_{ij}: 11:1, 14:3, 15:1, 21:3, 23:2, 34:3, 35:1, 38:2, 41:3, 48:1, 53:3, 54:1, 56:2, 64:2, 77:1$   
 $w_i \quad \langle 150t^4 - 162t^3 + 264t^2 - 576t + 432, \\ 42t^4 + 18t^3 + 48 \rangle$

$i = 61, D_i = 1, \rho_{ij}: 1:3, 10:3, 12:3, 25:1, 29:3, 36:3, 41:1, 55:1, 62:3, 64:1$   
 $w_i \quad \langle 32t^4 - 50t^2 - 2t + 44, \\ 4t^4 + 50t^3 + 10t^2 - 34t + 8 \rangle$

$i = 62, D_i = 1, e_i = 2, \rho_{ij}: 1:3, 2:2, 7:2, 8:1, 10:3, 15:3, 24:1, 26:1, 27:2, 28:3, 30:1, 32:2, 34:3, 35:2, 36:1, 37:1, 40:1, 45:3, 53:2, 57:3, 61:1, 67:2, 68:2, 70:2, 72:3, 76:3$   
 $w_i \quad \langle 315059456t^4 - 829089536t^3 + 1119233792t^2 - 67634432t + 184287744, \\ -96763648t^4 + 425905408t^3 - 124646656t^2 + 392540416t - 526420992 \rangle$

$i = 63, D_i = 1, \rho_{ij}: 5:1, 7:1, 12:3, 13:2, 15:3, 20:2, 30:2, 31:2, 38:3, 47:3, 54:3, 57:3, 72:1$   
 $w_i \quad \langle 32t^4 - 62t^3 - 26t^2 + 232t + 140, \\ 96t^4 + 174t^3 + 122t^2 - 204t - 280 \rangle$

$i = 64, D_i = 1, \rho_{ij}: 6:1, 9:2, 12:3, 14:3, 16:3, 17:3, 18:1, 21:3, 25:3, 26:2, 28:1, 33:2, 34:3, 35:2, 36:1, 37:2, 38:2, 39:1, 42:2, 46:2, 51:2, 56:2, 57:1, 60:2, 61:3, 65:1, 66:3, 72:3, 78:2$   
 $w_i \quad \langle 113780520t^2 - 368752356t - 168969492, \\ 305746956t^2 - 471938364t - 553130352 \rangle$

$i = 65, D_i = 1, \rho_{ij}: 3:1, 7:2, 15:2, 16:1, 18:2, 20:1, 21:2, 26:2, 29:3, 32:3, 36:3, 38:2, 45:2, 48:2, 53:1, 58:3, 64:3, 67:3, 72:1, 76:2$   
 $w_i \quad \langle 38862t^4 + 21346t^3 - 59310t^2 + 60986t + 34344, \\ -5062t^4 - 85078t^3 - 62298t^2 + 22022t - 47188 \rangle$

---

---

|                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $i = 66, D_i = 2, e_i = 1, \rho_{ij}: 1:1, 5:1, 6:2, 12:2, 14:3, 20:1, 21:3, 23:1, 26:3, 31:1, 33:2, 36:3, 37:1, 38:2, 41:2, 45:2, 48:2, 50:3, 59:3, 64:1, 73:1, 74:1, 81:2$ |
| $w_i \langle -1449856t^4 + 13937024t^3 + 14363008t^2 - 8466304t + 5814784, -7539328t^4 - 9792128t^3 + 6704000t^2 - 9270144t - 9115392 \rangle$                               |
| $i = 67, D_i = 1, \rho_{ij}: 0:3, 4:2, 5:1, 17:1, 26:3, 28:1, 38:3, 51:2, 58:3, 62:2, 65:1, 77:3$                                                                            |
| $w_i \langle 42t^4 - 30t^3 + 102t^2 - 78t + 84, -12t^4 + 24t^3 - 60t^2 + 132t - 12 \rangle$                                                                                  |
| $i = 68, D_i = 1, \rho_{ij}: 3:2, 16:3, 20:3, 22:2, 24:3, 26:1, 32:2, 36:3, 58:1, 62:2$                                                                                      |
| $w_i \langle 16t^4 + 30t^3 + 20t^2 - 40t + 24, 6t^4 + 6t^3 - 24t^2 + 4t + 28 \rangle$                                                                                        |
| $i = 69, D_i = 1, \rho_{ij}: 0:3, 2:3, 10:1, 11:2, 13:2, 14:1, 15:3, 16:1, 21:2, 25:1, 26:2, 27:3, 33:2, 36:3, 49:3, 70:3$                                                   |
| $w_i \langle 196t^4 - 34t^3 - 144t^2 + 350t + 828, 52t^4 - 178t^3 + 212t^2 + 490t - 804 \rangle$                                                                             |
| $i = 70, D_i = 1, \rho_{ij}: 8:2, 10:3, 11:3, 19:2, 24:3, 31:2, 35:1, 41:1, 42:3, 46:2, 48:3, 50:1, 57:2, 62:2, 69:1$                                                        |
| $w_i \langle 990t^4 + 11294t^3 - 17578t^2 - 9520t + 16388, -4918t^4 + 5172t^3 + 18910t^2 - 31300t + 11516 \rangle$                                                           |
| $i = 71, D_i = 1, \rho_{ij}: 2:3, 13:2, 18:2, 25:2, 26:3, 27:3, 28:2, 29:2, 52:3, 53:3, 54:1, 77:3$                                                                          |
| $w_i \langle -19t^4 + t^3 - 9t^2 + 20t + 82, 67t^4 + 33t^3 + 35t^2 + 54t - 62 \rangle$                                                                                       |
| $i = 72, D_i = 1, \rho_{ij}: 5:1, 9:2, 10:2, 12:3, 28:3, 29:1, 57:1, 62:1, 63:3, 64:1, 65:3$                                                                                 |
| $w_i \langle 2t^4 + 76t^3 - 86t^2 - 288t + 292, 34t^4 + 12t^3 - 274t^2 + 376t - 44 \rangle$                                                                                  |
| $i = 73, D_i = 1, \rho_{ij}: 0:2, 2:2, 3:3, 4:2, 15:1, 19:2, 24:3, 26:1, 30:2, 32:1, 36:3, 42:3, 50:1, 55:2, 66:3, 77:1$                                                     |
| $w_i \langle 16t^4 - 34t^3 + 14t^2 - 18t + 4, 20t^4 + 2t^3 + 28t^2 - 4t + 34 \rangle$                                                                                        |
| $i = 74, D_i = 1, \rho_{ij}: 3:2, 38:1, 49:3, 51:1, 58:1, 66:3, 76:3, 77:2$                                                                                                  |
| $w_i \langle 2t^4 + 12t^3 + 22t^2 + 16t + 8, -2t^4 - 6t^3 - 2t^2 + 6t + 4 \rangle$                                                                                           |
| $i = 75, D_i = 1, \rho_{ij}: 1:3, 2:1, 4:2, 7:3, 18:1, 27:1, 31:1, 32:3, 38:1$                                                                                               |
| $w_i \langle 2t^4 - 2t^3 + 3t^2 + 20t + 7, -2t^4 + t^3 + 17t^2 + 8t - 6 \rangle$                                                                                             |
| $i = 76, D_i = 1, \rho_{ij}: 11:3, 18:3, 24:1, 27:3, 37:1, 44:2, 45:3, 55:1, 62:1, 65:2, 74:1$                                                                               |
| $w_i \langle 40t^4 + 8t^3 - 88t^2 - 62t + 66, 10t^4 - 96t^3 + 60t^2 + 44t + 18 \rangle$                                                                                      |
| $i = 77, D_i = 1, \rho_{ij}: 1:1, 7:1, 10:1, 12:3, 18:1, 21:3, 24:1, 25:3, 34:2, 35:1, 41:1, 42:1, 45:3, 51:1, 55:2, 56:1, 59:2, 60:3, 67:1, 71:1, 73:3, 74:2, 81:2$         |
| $w_i \langle -313t^4 + 212t^3 + 684t^2 + 1361t + 390, -457t^4 - 720t^3 - 688t^2 - 221t - 8 \rangle$                                                                          |
| $i = 78, D_i = 1, \rho_{ij}: 7:1, 10:2, 16:3, 25:3, 30:3, 31:1, 32:3, 34:1, 37:2, 43:1, 46:3, 49:1, 53:3, 64:2$                                                              |
| $w_i \langle -114t^4 - 275t^3 + 268t^2 + 141t + 52, 78t^4 - 235t^3 - 186t^2 + 203t - 64 \rangle$                                                                             |
| $i = 79, D_i = 1, \rho_{ij}: 0:2, 8:2, 16:1, 22:2, 26:1, 27:3, 35:2, 36:2, 38:3, 39:2, 48:3, 51:1, 55:2$                                                                     |
| $w_i \langle 25t^4 - 42t^3 + 350t^2 - 222t + 36, t^4 - 144t^3 + 104t^2 + 162t - 72 \rangle$                                                                                  |
| $i = 80, D_i = 2, e_i = 0, \rho_{ij}: 1:2, 3:1, 6:3, 8:2, 10:1, 21:3, 22:2, 25:2, 27:1, 30:1, 32:1, 38:3, 39:1, 41:3, 48:2, 53:1$                                            |
| $w_i \langle 20544t^4 - 4896t^3 - 5952t^2 - 11584t + 352, -10880t^4 - 4576t^3 - 28480t^2 + 4480t + 1824 \rangle$                                                             |
| $i = 81, D_i = 1, \rho_{ij}: 4:2, 17:2, 34:2, 38:2, 51:2, 57:2, 66:2, 77:2$                                                                                                  |
| $w_i \langle 6t^3 + 18t^2 - 12t + 12, -6t^4 + 12t^3 - 12t^2 + 6t + 12 \rangle$                                                                                               |

---

Table 3: Edges  $(i, j)$  with a linear endpoint, the linear endpoint  $k$  used, and the norm of  $R_{ij} = \text{Res}_t(f_k, g_{kl})$ .

| $(i, j)$  | $k$ | $N(R_{ij})$             | $(i, j)$   | $k$ | $N(R_{ij})$                        |
|-----------|-----|-------------------------|------------|-----|------------------------------------|
| $(0, 9)$  | 0   | $2^3 \cdot 3^2 \cdot 5$ | $(23, 57)$ | 23  | $2^9 \cdot 5$                      |
| $(0, 13)$ | 0   | $2^3 \cdot 3^2 \cdot 5$ | $(23, 59)$ | 23  | $2^4 \cdot 5^2$                    |
| $(0, 16)$ | 0   | $2^6$                   | $(23, 60)$ | 23  | $2^4 \cdot 3^4$                    |
| $(0, 23)$ | 0   | $2^2 \cdot 3^2$         | $(23, 66)$ | 23  | $2^{10} \cdot 3^2 \cdot 5^2$       |
| $(0, 31)$ | 0   | $2 \cdot 5$             | $(24, 26)$ | 24  | $2^8 \cdot 3^2 \cdot 5^2 \cdot 17$ |
| $(0, 39)$ | 0   | $3^4$                   | $(24, 34)$ | 24  | $2^{10} \cdot 5^2$                 |
| $(0, 41)$ | 0   | $2^3 \cdot 3^2 \cdot 5$ | $(24, 36)$ | 24  | $2^{10} \cdot 5$                   |
| $(0, 55)$ | 0   | $2^2 \cdot 5^2$         | $(24, 37)$ | 24  | $2^{12} \cdot 5^2$                 |
| $(0, 67)$ | 0   | $2^3 \cdot 3^2$         | $(24, 39)$ | 24  | $2^8 \cdot 3^4 \cdot 5^5$          |

| $(i, j)$ | $k$ | $N(R_{ij})$             | $(i, j)$ | $k$ | $N(R_{ij})$                        |
|----------|-----|-------------------------|----------|-----|------------------------------------|
| (0, 69)  | 0   | $2^2 \cdot 5^2$         | (24, 44) | 24  | $2^2$                              |
| (0, 73)  | 0   | 1                       | (24, 55) | 24  | $2^8 \cdot 5^4$                    |
| (0, 79)  | 0   | $2^2 \cdot 3^2$         | (24, 59) | 24  | $2^5 \cdot 3^2 \cdot 17$           |
| (1, 14)  | 1   | $2^4 \cdot 17$          | (24, 62) | 24  | $2^{12} \cdot 17$                  |
| (1, 18)  | 1   | $2 \cdot 13^2$          | (24, 68) | 24  | $2^5 \cdot 5 \cdot 17$             |
| (1, 22)  | 1   | $2^6 \cdot 13^2$        | (24, 70) | 24  | $2^7 \cdot 17$                     |
| (1, 25)  | 1   | $2^6 \cdot 29$          | (24, 73) | 24  | $2^2 \cdot 17 \cdot 53$            |
| (1, 26)  | 1   | $2^6$                   | (24, 76) | 24  | $2^6 \cdot 3^2 \cdot 5$            |
| (1, 29)  | 1   | $2^5 \cdot 13 \cdot 17$ | (24, 77) | 24  | $2 \cdot 5^2 \cdot 53$             |
| (1, 31)  | 1   | $2^5 \cdot 13 \cdot 17$ | (25, 31) | 25  | $2^7 \cdot 5^2 \cdot 53$           |
| (1, 35)  | 1   | $2^8$                   | (25, 32) | 25  | $2^8$                              |
| (1, 37)  | 1   | $2^8$                   | (25, 35) | 25  | $2^{10} \cdot 5^2$                 |
| (1, 41)  | 1   | $2^5 \cdot 17^2$        | (25, 39) | 25  | $2^8 \cdot 3^4 \cdot 5^5$          |
| (1, 48)  | 1   | $2^2 \cdot 13^2$        | (25, 45) | 25  | $2^2 \cdot 3^4 \cdot 5^4$          |
| (1, 55)  | 1   | $2^6 \cdot 13 \cdot 17$ | (25, 47) | 25  | $2^6 \cdot 5^2$                    |
| (1, 61)  | 1   | $2 \cdot 13$            | (25, 52) | 25  | $2^7 \cdot 5^3 \cdot 17$           |
| (1, 62)  | 1   | $2^{14}$                | (25, 54) | 25  | $2^9 \cdot 3^2 \cdot 5$            |
| (1, 66)  | 1   | $2^{14}$                | (25, 61) | 25  | $2^4 \cdot 5$                      |
| (1, 75)  | 1   | $2^4 \cdot 13 \cdot 17$ | (25, 64) | 25  | $2^2 \cdot 5^4$                    |
| (1, 77)  | 1   | $2^4$                   | (25, 69) | 25  | $2^4 \cdot 5^2 \cdot 11^2$         |
| (1, 80)  | 1   | $2^5 \cdot 13 \cdot 29$ | (25, 71) | 25  | $2^2 \cdot 17 \cdot 53$            |
| (2, 8)   | 2   | $2^3 \cdot 3^2 \cdot 5$ | (25, 77) | 25  | $2 \cdot 5^2 \cdot 53$             |
| (2, 12)  | 2   | $17^2$                  | (25, 78) | 25  | $2^4 \cdot 3^4$                    |
| (2, 17)  | 2   | $3^2 \cdot 5$           | (25, 80) | 25  | $2^{14} \cdot 5 \cdot 29$          |
| (2, 18)  | 2   | $3^2 \cdot 5$           | (26, 27) | 26  | $2^7 \cdot 5 \cdot 29$             |
| (2, 20)  | 2   | 1                       | (26, 28) | 26  | $2^6 \cdot 13^2 \cdot 17$          |
| (2, 21)  | 2   | $3^2 \cdot 5^2$         | (26, 29) | 26  | $2^8 \cdot 5^2 \cdot 17$           |
| (2, 25)  | 2   | $2 \cdot 5$             | (26, 31) | 26  | $2^8 \cdot 5^2 \cdot 17$           |
| (2, 26)  | 2   | $2^6 \cdot 5^2$         | (26, 34) | 26  | $2^9 \cdot 5$                      |
| (2, 33)  | 2   | $2 \cdot 5$             | (26, 35) | 26  | $2^9 \cdot 5$                      |
| (2, 34)  | 2   | $2 \cdot 3^2 \cdot 17$  | (26, 36) | 26  | $2^{16}$                           |
| (2, 35)  | 2   | $2 \cdot 3^2 \cdot 17$  | (26, 41) | 26  | $2^8 \cdot 5 \cdot 17^2$           |
| (2, 38)  | 2   | $2^4$                   | (26, 43) | 26  | $2^8 \cdot 5^2 \cdot 29$           |
| (2, 39)  | 2   | $5^4$                   | (26, 44) | 26  | $2^4$                              |
| (2, 45)  | 2   | $3^2 \cdot 5^2$         | (26, 46) | 26  | $2^4 \cdot 3^2 \cdot 5$            |
| (2, 52)  | 2   | $2 \cdot 5$             | (26, 54) | 26  | $2^{10} \cdot 3^2 \cdot 5$         |
| (2, 62)  | 2   | $2^5 \cdot 5^2$         | (26, 55) | 26  | $2^{15} \cdot 5^2$                 |
| (2, 69)  | 2   | $2^2 \cdot 5^2$         | (26, 58) | 26  | $2^4 \cdot 5 \cdot 17$             |
| (2, 71)  | 2   | 1                       | (26, 59) | 26  | $2^4 \cdot 5 \cdot 17$             |
| (2, 73)  | 2   | 1                       | (26, 62) | 26  | $2^{10} \cdot 3^2 \cdot 5$         |
| (2, 75)  | 2   | $2^2 \cdot 3^2$         | (26, 64) | 26  | $2^2 \cdot 5 \cdot 29$             |
| (3, 7)   | 3   | $3^2 \cdot 5$           | (26, 65) | 26  | $2^4 \cdot 17$                     |
| (3, 9)   | 3   | $2^3 \cdot 3^2 \cdot 5$ | (26, 66) | 26  | $2^{10} \cdot 17$                  |
| (3, 16)  | 3   | $3^4$                   | (26, 67) | 26  | $2^5 \cdot 13^2 \cdot 17$          |
| (3, 25)  | 3   | $2^2 \cdot 3^2$         | (26, 68) | 26  | $2^3 \cdot 5^2$                    |
| (3, 32)  | 3   | $2 \cdot 5$             | (26, 69) | 26  | $2^4 \cdot 5^2 \cdot 17$           |
| (3, 34)  | 3   | $2^2 \cdot 5^2$         | (26, 71) | 26  | $2^6 \cdot 17$                     |
| (3, 38)  | 3   | $2 \cdot 5$             | (26, 73) | 26  | $2^6 \cdot 17$                     |
| (3, 40)  | 3   | $2^3 \cdot 3^2 \cdot 5$ | (26, 79) | 26  | $2^4 \cdot 3^2 \cdot 5$            |
| (3, 41)  | 3   | $2^3 \cdot 3^2 \cdot 5$ | (27, 29) | 27  | $2^7 \cdot 5^3 \cdot 17$           |
| (3, 54)  | 3   | $2^3 \cdot 3^2$         | (27, 31) | 27  | $2^7 \cdot 5^4$                    |
| (3, 57)  | 3   | $2^2 \cdot 3^2$         | (27, 38) | 27  | $2^7 \cdot 5^2$                    |
| (3, 58)  | 3   | $3^2 \cdot 5^2$         | (27, 43) | 27  | $2^8 \cdot 3^2 \cdot 5^3 \cdot 29$ |
| (3, 65)  | 3   | $2^3 \cdot 3^2$         | (27, 45) | 27  | $2^2 \cdot 5^6$                    |
| (3, 68)  | 3   | 1                       | (27, 46) | 27  | $2^3 \cdot 5^3$                    |
| (3, 73)  | 3   | 1                       | (27, 50) | 27  | $2^4 \cdot 3^2$                    |
| (3, 74)  | 3   | $2^2 \cdot 5^2$         | (27, 55) | 27  | $2^7 \cdot 5^3 \cdot 17$           |

| $(i, j)$ | $k$ | $N(R_{ij})$                        | $(i, j)$ | $k$ | $N(R_{ij})$                                 |
|----------|-----|------------------------------------|----------|-----|---------------------------------------------|
| (3, 80)  | 3   | $2^3 \cdot 3^2 \cdot 5$            | (27, 56) | 27  | 2                                           |
| (4, 6)   | 4   | $3^2 \cdot 5$                      | (27, 59) | 27  | $2^2 \cdot 5$                               |
| (4, 7)   | 4   | $3^2 \cdot 5$                      | (27, 62) | 27  | $2^9 \cdot 5^4$                             |
| (4, 16)  | 4   | $3^4$                              | (27, 69) | 27  | $2^3 \cdot 5^3$                             |
| (4, 20)  | 4   | $3^2 \cdot 5^2$                    | (27, 71) | 27  | $2^2$                                       |
| (4, 25)  | 4   | $2^6 \cdot 5^2$                    | (27, 75) | 27  | $2^4 \cdot 3^2 \cdot 5$                     |
| (4, 28)  | 4   | $2^3 \cdot 3^2$                    | (27, 76) | 27  | $2^7 \cdot 5^3$                             |
| (4, 29)  | 4   | $2^3 \cdot 3^2$                    | (27, 79) | 27  | $2^2 \cdot 3^4 \cdot 5$                     |
| (4, 31)  | 4   | $2^2 \cdot 5^2$                    | (27, 80) | 27  | $2^{10} \cdot 5^2$                          |
| (4, 35)  | 4   | $2^2 \cdot 5^2$                    | (28, 31) | 28  | $2^8 \cdot 5^2 \cdot 17$                    |
| (4, 38)  | 4   | $2 \cdot 5$                        | (28, 32) | 28  | $2^8 \cdot 13$                              |
| (4, 42)  | 4   | $2^3 \cdot 3^2 \cdot 5$            | (28, 46) | 28  | $2^4 \cdot 3^2 \cdot 5$                     |
| (4, 44)  | 4   | 1                                  | (28, 50) | 28  | $2^2 \cdot 5^2 \cdot 37$                    |
| (4, 57)  | 4   | $2^3 \cdot 3^2$                    | (28, 51) | 28  | $2 \cdot 3^6$                               |
| (4, 59)  | 4   | $3^2 \cdot 5^2$                    | (28, 54) | 28  | $2^7 \cdot 17$                              |
| (4, 67)  | 4   | $2^3 \cdot 3^2$                    | (28, 62) | 28  | $2^{16} \cdot 13$                           |
| (4, 73)  | 4   | 1                                  | (28, 64) | 28  | $2 \cdot 3^2 \cdot 17 \cdot 37$             |
| (4, 75)  | 4   | $2 \cdot 5$                        | (28, 67) | 28  | $2^4 \cdot 3^2 \cdot 13^2 \cdot 17$         |
| (4, 81)  | 4   | $2^3 \cdot 3^2$                    | (28, 71) | 28  | $2^2 \cdot 17$                              |
| (5, 9)   | 5   | $2^3 \cdot 3^2 \cdot 5$            | (28, 72) | 28  | $2^2 \cdot 3^4 \cdot 5^2 \cdot 13 \cdot 17$ |
| (5, 12)  | 5   | 1                                  | (29, 32) | 29  | $2^7 \cdot 5^4$                             |
| (5, 13)  | 5   | $2^3 \cdot 5^4$                    | (29, 41) | 29  | $2^8 \cdot 5^3 \cdot 13 \cdot 17$           |
| (5, 14)  | 5   | $2 \cdot 5$                        | (29, 45) | 29  | $2^4 \cdot 5^6$                             |
| (5, 16)  | 5   | $5^4$                              | (29, 47) | 29  | $2^2 \cdot 5^2 \cdot 17$                    |
| (5, 17)  | 5   | $3^2 \cdot 5$                      | (29, 50) | 29  | $2^2 \cdot 5^2 \cdot 13$                    |
| (5, 25)  | 5   | $2 \cdot 5$                        | (29, 55) | 29  | $2^8 \cdot 5^3 \cdot 17^2$                  |
| (5, 26)  | 5   | $2^2 \cdot 3^2$                    | (29, 58) | 29  | $2^2 \cdot 5^3 \cdot 13$                    |
| (5, 27)  | 5   | $2^3 \cdot 3^2$                    | (29, 61) | 29  | $2^2 \cdot 3^2 \cdot 17$                    |
| (5, 35)  | 5   | $2 \cdot 5$                        | (29, 65) | 29  | $2^7 \cdot 5^4$                             |
| (5, 36)  | 5   | $2^4$                              | (29, 71) | 29  | $2^4$                                       |
| (5, 39)  | 5   | $3^4$                              | (29, 72) | 29  | $2^2 \cdot 5^4$                             |
| (5, 63)  | 5   | 1                                  | (30, 31) | 30  | $2^7 \cdot 5^3 \cdot 17$                    |
| (5, 66)  | 5   | $2^5 \cdot 5^2$                    | (30, 33) | 30  | $2^8 \cdot 5$                               |
| (5, 67)  | 5   | $2^3 \cdot 3^2$                    | (30, 41) | 30  | $2^8 \cdot 5^6$                             |
| (5, 72)  | 5   | 1                                  | (30, 46) | 30  | $2^3 \cdot 5^4$                             |
| (6, 10)  | 6   | $5^3$                              | (30, 47) | 30  | $2^2 \cdot 3^2$                             |
| (6, 11)  | 6   | 5                                  | (30, 50) | 30  | $2^4 \cdot 5^2$                             |
| (6, 14)  | 6   | $5^3 \cdot 13$                     | (30, 52) | 30  | $2^8 \cdot 5 \cdot 41$                      |
| (6, 20)  | 6   | $5^2 \cdot 17$                     | (30, 62) | 30  | $2^9 \cdot 5^3$                             |
| (6, 25)  | 6   | $2^2 \cdot 5^3 \cdot 17$           | (30, 63) | 30  | $2^2 \cdot 5^3 \cdot 17$                    |
| (6, 28)  | 6   | $2^4 \cdot 3^2 \cdot 13$           | (30, 73) | 30  | $2^2$                                       |
| (6, 30)  | 6   | $2^2 \cdot 3^2 \cdot 5^2 \cdot 41$ | (30, 78) | 30  | $2 \cdot 5^2$                               |
| (6, 32)  | 6   | $2^2 \cdot 5^2 \cdot 13$           | (30, 80) | 30  | $2^{10} \cdot 3^2$                          |
| (6, 33)  | 6   | $2^4 \cdot 3^4 \cdot 5^2$          | (31, 37) | 31  | $2^6 \cdot 3^2 \cdot 5^5$                   |
| (6, 35)  | 6   | $2^2 \cdot 5^3 \cdot 29$           | (31, 41) | 31  | $2^8 \cdot 5^6 \cdot 17$                    |
| (6, 40)  | 6   | $2^7 \cdot 3^4 \cdot 5^3$          | (31, 42) | 31  | $2^{10} \cdot 5^6$                          |
| (6, 41)  | 6   | $2^7 \cdot 3^2 \cdot 5^3 \cdot 13$ | (31, 53) | 31  | $2 \cdot 3^2 \cdot 5^2 \cdot 53$            |
| (6, 42)  | 6   | $2^9 \cdot 3^4 \cdot 5^2$          | (31, 63) | 31  | $2^2 \cdot 3^2 \cdot 17$                    |
| (6, 48)  | 6   | $3^2 \cdot 5^3$                    | (31, 66) | 31  | $2^{13} \cdot 5^4$                          |
| (6, 51)  | 6   | $3^4 \cdot 5^2$                    | (31, 70) | 31  | $2^2 \cdot 5^4$                             |
| (6, 52)  | 6   | $2^4 \cdot 5 \cdot 41$             | (31, 75) | 31  | $2 \cdot 5^3 \cdot 17$                      |
| (6, 53)  | 6   | $3^2 \cdot 5 \cdot 29$             | (31, 78) | 31  | $2^2 \cdot 5^2$                             |
| (6, 56)  | 6   | $3^4 \cdot 13$                     | (32, 38) | 32  | $2^7 \cdot 17^2$                            |
| (6, 64)  | 6   | $3^2 \cdot 5 \cdot 13$             | (32, 44) | 32  | $2^2$                                       |
| (6, 66)  | 6   | $2^5 \cdot 3^2 \cdot 5^3$          | (32, 49) | 32  | $2 \cdot 5 \cdot 17 \cdot 29$               |
| (6, 80)  | 6   | $2^{10} \cdot 3^2 \cdot 5$         | (32, 50) | 32  | $2^8 \cdot 5^2$                             |
| (7, 11)  | 7   | $5^3$                              | (32, 62) | 32  | $2^9 \cdot 5 \cdot 13 \cdot 41$             |

| $(i, j)$ | $k$ | $N(R_{ij})$                                    | $(i, j)$ | $k$ | $N(R_{ij})$                               |
|----------|-----|------------------------------------------------|----------|-----|-------------------------------------------|
| (7, 28)  | 7   | $2^6 \cdot 3^2 \cdot 5^2$                      | (32, 65) | 32  | $2^3 \cdot 3^4 \cdot 17^2$                |
| (7, 29)  | 7   | $2^2 \cdot 3^4 \cdot 5^3$                      | (32, 68) | 32  | $2^2 \cdot 41^2$                          |
| (7, 32)  | 7   | $2^2 \cdot 17 \cdot 41$                        | (32, 73) | 32  | $2^2 \cdot 29$                            |
| (7, 35)  | 7   | $2^2 \cdot 3^4 \cdot 17$                       | (32, 75) | 32  | $2^3 \cdot 3^2 \cdot 5 \cdot 17 \cdot 37$ |
| (7, 38)  | 7   | $2^2 \cdot 5 \cdot 17$                         | (32, 78) | 32  | $2 \cdot 5^2 \cdot 29$                    |
| (7, 39)  | 7   | $2^6 \cdot 5^3 \cdot 41$                       | (32, 80) | 32  | $2^{10} \cdot 17 \cdot 37$                |
| (7, 40)  | 7   | $2^7 \cdot 3^4 \cdot 5 \cdot 17$               | (33, 36) | 33  | $2^6 \cdot 13 \cdot 17$                   |
| (7, 41)  | 7   | $2^7 \cdot 3^4 \cdot 5^4$                      | (33, 53) | 33  | $2 \cdot 3^4 \cdot 5 \cdot 13$            |
| (7, 42)  | 7   | $2^9 \cdot 3^4 \cdot 5^2$                      | (33, 64) | 33  | $2 \cdot 3^2 \cdot 17^2$                  |
| (7, 43)  | 7   | $2^7 \cdot 3^2 \cdot 5^3$                      | (33, 66) | 33  | $2^{10} \cdot 3^2 \cdot 5$                |
| (7, 48)  | 7   | $3^2 \cdot 5^3$                                | (33, 69) | 33  | $2^3 \cdot 5^2 \cdot 13^2 \cdot 17$       |
| (7, 50)  | 7   | $2^5 \cdot 3^2 \cdot 5$                        | (34, 38) | 34  | $2^{13} \cdot 5$                          |
| (7, 53)  | 7   | $3^2 \cdot 5 \cdot 13$                         | (34, 42) | 34  | $2^{14} \cdot 5 \cdot 17$                 |
| (7, 55)  | 7   | $2^2 \cdot 5^4$                                | (34, 44) | 34  | $2^2$                                     |
| (7, 57)  | 7   | $2^2 \cdot 3^4 \cdot 13 \cdot 61$              | (34, 48) | 34  | $2^5 \cdot 3^2 \cdot 5$                   |
| (7, 62)  | 7   | $2^5 \cdot 5 \cdot 13 \cdot 17$                | (34, 51) | 34  | $2^5 \cdot 3^2 \cdot 5$                   |
| (7, 63)  | 7   | $5 \cdot 13 \cdot 61$                          | (34, 53) | 34  | $2^7 \cdot 5$                             |
| (7, 65)  | 7   | $3^2 \cdot 5^3$                                | (34, 55) | 34  | $2^9 \cdot 5$                             |
| (7, 75)  | 7   | $3^2 \cdot 5 \cdot 17$                         | (34, 56) | 34  | $2^4$                                     |
| (7, 77)  | 7   | 5                                              | (34, 60) | 34  | $2^7 \cdot 41$                            |
| (7, 78)  | 7   | $5 \cdot 13$                                   | (34, 62) | 34  | $2^{13} \cdot 17$                         |
| (8, 11)  | 8   | $2^3 \cdot 5^2 \cdot 13$                       | (34, 64) | 34  | $2^6 \cdot 17 \cdot 29$                   |
| (8, 15)  | 8   | $2^{16} \cdot 5^3 \cdot 13$                    | (34, 77) | 34  | $2 \cdot 17$                              |
| (8, 32)  | 8   | $2^{16} \cdot 5^3 \cdot 17$                    | (34, 78) | 34  | $2^5 \cdot 5$                             |
| (8, 40)  | 8   | $2^{21} \cdot 3^4 \cdot 5^2$                   | (34, 81) | 34  | $2^7 \cdot 17$                            |
| (8, 41)  | 8   | $2^{21} \cdot 3^4 \cdot 5^2$                   | (35, 37) | 35  | $2^{10}$                                  |
| (8, 42)  | 8   | $2^{21} \cdot 3^4 \cdot 5^2$                   | (35, 44) | 35  | $2^2$                                     |
| (8, 48)  | 8   | $2^3 \cdot 3^4 \cdot 5 \cdot 17$               | (35, 45) | 35  | $2^2 \cdot 5^2 \cdot 17^2$                |
| (8, 49)  | 8   | $2^3 \cdot 3^2 \cdot 5 \cdot 13^2$             | (35, 46) | 35  | $2^7 \cdot 17$                            |
| (8, 50)  | 8   | $2^{11} \cdot 3^4 \cdot 5 \cdot 13$            | (35, 48) | 35  | $2^5 \cdot 3^2 \cdot 5$                   |
| (8, 57)  | 8   | $2^{16} \cdot 3^4 \cdot 5 \cdot 13^2 \cdot 17$ | (35, 53) | 35  | $2^6 \cdot 17 \cdot 29$                   |
| (8, 59)  | 8   | $2^{11} \cdot 5^2 \cdot 13$                    | (35, 60) | 35  | $2^7 \cdot 41$                            |
| (8, 62)  | 8   | $2^{16} \cdot 5$                               | (35, 62) | 35  | $2^{12} \cdot 17$                         |
| (8, 70)  | 8   | $2^{11} \cdot 3^2 \cdot 5^2 \cdot 13$          | (35, 64) | 35  | $2^7 \cdot 5$                             |
| (8, 79)  | 8   | $2^{11} \cdot 3^2 \cdot 5^2 \cdot 17$          | (35, 70) | 35  | $2^6 \cdot 3^2 \cdot 17$                  |
| (8, 80)  | 8   | $2^{16} \cdot 3^2 \cdot 5^2 \cdot 17$          | (35, 77) | 35  | $2 \cdot 17$                              |
| (9, 10)  | 9   | $2^3 \cdot 5^2 \cdot 13$                       | (35, 79) | 35  | $2^{10}$                                  |
| (9, 16)  | 9   | $2^{16} \cdot 5^3 \cdot 13$                    | (36, 37) | 36  | $2^{11} \cdot 5$                          |
| (9, 18)  | 9   | $2^{16} \cdot 3^4 \cdot 5 \cdot 37$            | (36, 45) | 36  | $2^8 \cdot 5^2$                           |
| (9, 21)  | 9   | $2^{16} \cdot 5^3 \cdot 13$                    | (36, 46) | 36  | $2^6 \cdot 13$                            |
| (9, 32)  | 9   | $2^{16} \cdot 37$                              | (36, 48) | 36  | $2^6$                                     |
| (9, 35)  | 9   | $2^{16} \cdot 37$                              | (36, 49) | 36  | $2^8 \cdot 5^4$                           |
| (9, 37)  | 9   | $2^{16} \cdot 3^2 \cdot 5^5$                   | (36, 53) | 36  | $2^5 \cdot 13 \cdot 17$                   |
| (9, 38)  | 9   | $2^{16} \cdot 5 \cdot 17 \cdot 37$             | (36, 57) | 36  | $2^{20}$                                  |
| (9, 41)  | 9   | $2^{21} \cdot 3^2 \cdot 5^3 \cdot 13$          | (36, 59) | 36  | $2^8 \cdot 5$                             |
| (9, 46)  | 9   | $2^{11} \cdot 5$                               | (36, 61) | 36  | $2^3 \cdot 5^2 \cdot 13^2$                |
| (9, 48)  | 9   | $2^3 \cdot 3^4 \cdot 5 \cdot 17$               | (36, 62) | 36  | $2^{12} \cdot 13$                         |
| (9, 49)  | 9   | $2^3 \cdot 3^2 \cdot 5 \cdot 13^2$             | (36, 64) | 36  | $2^5 \cdot 13 \cdot 17$                   |
| (9, 52)  | 9   | $2^{16} \cdot 5^6$                             | (36, 65) | 36  | $2^6 \cdot 17 \cdot 37$                   |
| (9, 58)  | 9   | $2^{11} \cdot 5^2 \cdot 13$                    | (36, 66) | 36  | $2^{12} \cdot 17 \cdot 37$                |
| (9, 64)  | 9   | $2^3 \cdot 3^2 \cdot 5^2 \cdot 37$             | (36, 68) | 36  | $2^3$                                     |
| (9, 72)  | 9   | $2^{11} \cdot 3^2 \cdot 5^2 \cdot 13$          | (36, 69) | 36  | $2^7 \cdot 5 \cdot 17$                    |
| (10, 14) | 10  | $13^2$                                         | (36, 73) | 36  | $2^6 \cdot 17$                            |
| (10, 15) | 10  | $5^2 \cdot 17$                                 | (36, 79) | 36  | $2^6 \cdot 17^2$                          |
| (10, 20) | 10  | $2 \cdot 5$                                    | (37, 42) | 37  | $2^{10} \cdot 5^3$                        |
| (10, 30) | 10  | $2 \cdot 5$                                    | (37, 45) | 37  | $2^2 \cdot 5^3 \cdot 53$                  |
| (10, 36) | 10  | $2^4 \cdot 5$                                  | (37, 47) | 37  | $2^5$                                     |



| $(i, j)$ | $k$ | $N(R_{ij})$                          | $(i, j)$ | $k$ | $N(R_{ij})$                           |
|----------|-----|--------------------------------------|----------|-----|---------------------------------------|
| (10, 41) | 10  | $2^3 \cdot 5^2 \cdot 13$             | (37, 49) | 37  | $2^3 \cdot 5 \cdot 13$                |
| (10, 45) | 10  | $5^2 \cdot 17$                       | (37, 50) | 37  | $2^2 \cdot 5^2 \cdot 37$              |
| (10, 51) | 10  | $2 \cdot 5^4$                        | (37, 53) | 37  | $2^3 \cdot 3^2 \cdot 5^2$             |
| (10, 61) | 10  | $2 \cdot 5$                          | (37, 62) | 37  | $2^{10} \cdot 3^2$                    |
| (10, 62) | 10  | $2^6 \cdot 17$                       | (37, 64) | 37  | $2^3 \cdot 37 \cdot 41$               |
| (10, 69) | 10  | $2 \cdot 5 \cdot 13$                 | (37, 66) | 37  | $2^{10} \cdot 5^4$                    |
| (10, 70) | 10  | $2^4 \cdot 5$                        | (37, 76) | 37  | $2^5 \cdot 53$                        |
| (10, 72) | 10  | $2^5 \cdot 13$                       | (37, 78) | 37  | $2 \cdot 5^2 \cdot 41$                |
| (10, 77) | 10  | 5                                    | (38, 44) | 38  | $2^4$                                 |
| (10, 78) | 10  | $5^3$                                | (38, 48) | 38  | $2^4 \cdot 5^2 \cdot 17^2$            |
| (10, 80) | 10  | $2^3 \cdot 13^2$                     | (38, 49) | 38  | $2^4 \cdot 17^2$                      |
| (11, 14) | 11  | $5^2 \cdot 17$                       | (38, 51) | 38  | $2^4 \cdot 5^2$                       |
| (11, 15) | 11  | $13^2$                               | (38, 54) | 38  | $2^9$                                 |
| (11, 41) | 11  | $2^3 \cdot 5^3 \cdot 13$             | (38, 56) | 38  | $2^4 \cdot 17^2$                      |
| (11, 45) | 11  | $5^2 \cdot 17$                       | (38, 60) | 38  | $2^6$                                 |
| (11, 54) | 11  | $2^2 \cdot 5$                        | (38, 63) | 38  | $2^3$                                 |
| (11, 60) | 11  | $2 \cdot 13$                         | (38, 64) | 38  | $2^4 \cdot 37$                        |
| (11, 69) | 11  | $2 \cdot 5$                          | (38, 65) | 38  | $2^5 \cdot 17$                        |
| (11, 70) | 11  | $2^5 \cdot 13$                       | (38, 66) | 38  | $2^{11} \cdot 17$                     |
| (11, 76) | 11  | $2^4 \cdot 5$                        | (38, 67) | 38  | $2^7 \cdot 17$                        |
| (12, 16) | 12  | $2^{14}$                             | (38, 74) | 38  | $2 \cdot 17^2$                        |
| (12, 18) | 12  | $2^4 \cdot 13 \cdot 17$              | (38, 75) | 38  | $2^2 \cdot 17^2$                      |
| (12, 25) | 12  | $2^6 \cdot 3^2 \cdot 5$              | (38, 79) | 38  | $2^6 \cdot 17$                        |
| (12, 31) | 12  | $2^2 \cdot 3^2 \cdot 5^2 \cdot 13$   | (38, 80) | 38  | $2^8 \cdot 5^3 \cdot 17$              |
| (12, 34) | 12  | $2^6 \cdot 17^2$                     | (38, 81) | 38  | $2^7 \cdot 17$                        |
| (12, 35) | 12  | $2^6 \cdot 17^2$                     | (39, 44) | 39  | $2^{20}$                              |
| (12, 38) | 12  | $2^6$                                | (39, 49) | 39  | $3^6$                                 |
| (12, 41) | 12  | $2^5 \cdot 5^4 \cdot 41$             | (39, 50) | 39  | $2^{20} \cdot 5^2 \cdot 37$           |
| (12, 48) | 12  | $2^2 \cdot 13^2$                     | (39, 52) | 39  | $2^{18} \cdot 5^4$                    |
| (12, 49) | 12  | $2^2 \cdot 5^2$                      | (39, 53) | 39  | $5^4 \cdot 37$                        |
| (12, 61) | 12  | $2^5 \cdot 3^2 \cdot 5$              | (39, 64) | 39  | $5^4 \cdot 37$                        |
| (12, 63) | 12  | $2^5 \cdot 3^2 \cdot 5$              | (39, 79) | 39  | $2^{12} \cdot 3^4 \cdot 5^5$          |
| (12, 64) | 12  | $2^2 \cdot 3^2 \cdot 17$             | (39, 80) | 39  | $2^{26} \cdot 5^2 \cdot 37$           |
| (12, 66) | 12  | $2^9 \cdot 41$                       | (40, 44) | 40  | $2^{20} \cdot 17$                     |
| (12, 72) | 12  | $2^{14} \cdot 5$                     | (40, 62) | 40  | $2^{24} \cdot 3^2 \cdot 5 \cdot 17^2$ |
| (12, 77) | 12  | $5^2 \cdot 17^2$                     | (41, 45) | 41  | $2^{20} \cdot 5^6$                    |
| (13, 19) | 13  | $2^{11} \cdot 5 \cdot 7^2 \cdot 13$  | (41, 59) | 41  | $2^{18} \cdot 5^2 \cdot 17$           |
| (13, 25) | 13  | $2^{13} \cdot 7^2 \cdot 11^2$        | (41, 60) | 41  | $2^{18} \cdot 13^4$                   |
| (13, 41) | 13  | $2^{18} \cdot 5 \cdot 13 \cdot 17^2$ | (41, 61) | 41  | $2^{18} \cdot 5^4 \cdot 13$           |
| (13, 63) | 13  | $2^8$                                | (41, 66) | 41  | $2^{24} \cdot 5^2 \cdot 17 \cdot 41$  |
| (13, 69) | 13  | $2^8 \cdot 5 \cdot 11^2$             | (41, 70) | 41  | $2^{18} \cdot 5^4$                    |
| (13, 71) | 13  | $2^{13}$                             | (41, 77) | 41  | $2^{18} \cdot 5^4$                    |
| (14, 26) | 14  | $2 \cdot 5^3 \cdot 17$               | (41, 80) | 41  | $2^{26} \cdot 3^2 \cdot 13$           |
| (14, 35) | 14  | $2 \cdot 5 \cdot 17 \cdot 41$        | (42, 45) | 42  | $2^{22} \cdot 5^6$                    |
| (14, 41) | 14  | $2^6 \cdot 5^2 \cdot 13^2 \cdot 17$  | (42, 50) | 42  | $2^{22} \cdot 3^2 \cdot 5 \cdot 41$   |
| (14, 45) | 14  | $5^5 \cdot 17$                       | (42, 56) | 42  | $2^{18} \cdot 3^6 \cdot 11^2$         |
| (14, 46) | 14  | $2^3 \cdot 13 \cdot 17$              | (42, 64) | 42  | $2^3 \cdot 3^2 \cdot 11^2 \cdot 17$   |
| (14, 52) | 14  | $2 \cdot 5^3 \cdot 17$               | (42, 70) | 42  | $2^{18} \cdot 3^4 \cdot 5^5$          |
| (14, 60) | 14  | $2^5 \cdot 41$                       | (42, 73) | 42  | $2^{22} \cdot 17 \cdot 29 \cdot 41$   |
| (14, 64) | 14  | $5^2 \cdot 13$                       | (42, 77) | 42  | $2^{18} \cdot 5 \cdot 17^2$           |
| (14, 66) | 14  | $2^9 \cdot 5^2$                      | (43, 52) | 43  | $2^{20} \cdot 5^4$                    |
| (14, 69) | 14  | $2^5 \cdot 5^2 \cdot 13$             | (43, 78) | 43  | $2^{18} \cdot 29^2$                   |
| (15, 24) | 15  | $2 \cdot 5^3 \cdot 17$               | (44, 50) | 44  | $2^5$                                 |
| (15, 29) | 15  | $2 \cdot 5$                          | (44, 55) | 44  | $2^6 \cdot 17$                        |
| (15, 30) | 15  | $2 \cdot 5$                          | (44, 56) | 44  | 1                                     |
| (15, 34) | 15  | $2 \cdot 5 \cdot 17 \cdot 41$        | (44, 59) | 44  | 1                                     |
| (15, 45) | 15  | $5^5 \cdot 17$                       | (44, 76) | 44  | 17                                    |

| $(i, j)$ | $k$ | $N(R_{ij})$                             | $(i, j)$ | $k$ | $N(R_{ij})$                        |
|----------|-----|-----------------------------------------|----------|-----|------------------------------------|
| (15, 46) | 15  | $2^3 \cdot 5^3$                         | (45, 48) | 45  | $3^2 \cdot 5^2$                    |
| (15, 53) | 15  | $5^2 \cdot 13$                          | (45, 50) | 45  | $2^5 \cdot 5^3$                    |
| (15, 60) | 15  | $2^5 \cdot 41$                          | (45, 51) | 45  | $3^2 \cdot 5^6$                    |
| (15, 62) | 15  | $2^9 \cdot 13 \cdot 17$                 | (45, 62) | 45  | $2^5 \cdot 3^2 \cdot 5^3 \cdot 17$ |
| (15, 63) | 15  | $5^2 \cdot 13$                          | (45, 65) | 45  | $3^2 \cdot 5^3$                    |
| (15, 65) | 15  | $2^3 \cdot 5^2$                         | (45, 66) | 45  | $2^5 \cdot 3^2 \cdot 5^3$          |
| (15, 69) | 15  | $2^5 \cdot 5^2 \cdot 17$                | (45, 76) | 45  | $3^2 \cdot 5^2 \cdot 53$           |
| (15, 73) | 15  | $17^2$                                  | (45, 77) | 45  | $3^4 \cdot 5^2$                    |
| (16, 20) | 16  | $2^{14} \cdot 3^4$                      | (46, 49) | 46  | $2 \cdot 3^2 \cdot 29$             |
| (16, 21) | 16  | $2^{14} \cdot 5^6$                      | (46, 54) | 46  | $2^9 \cdot 5^2$                    |
| (16, 23) | 16  | $2^{16} \cdot 5^4$                      | (46, 57) | 46  | $2^8 \cdot 3^2 \cdot 13$           |
| (16, 25) | 16  | $2^{18} \cdot 3^2 \cdot 5^3 \cdot 11^2$ | (46, 64) | 46  | $2 \cdot 3^4 \cdot 13$             |
| (16, 36) | 16  | $2^{20}$                                | (46, 70) | 46  | $2^5 \cdot 13$                     |
| (16, 37) | 16  | $2^{16} \cdot 5^4 \cdot 13$             | (46, 78) | 46  | $2^3 \cdot 5 \cdot 13 \cdot 29$    |
| (16, 49) | 16  | $2^6 \cdot 13^2$                        | (47, 63) | 47  | $2^6 \cdot 5$                      |
| (16, 58) | 16  | $2^{18} \cdot 3^2 \cdot 13$             | (48, 50) | 48  | $5 \cdot 13 \cdot 17$              |
| (16, 59) | 16  | $2^{18} \cdot 3^2 \cdot 13$             | (48, 60) | 48  | $2^4 \cdot 3^4$                    |
| (16, 64) | 16  | $2^6 \cdot 3^2 \cdot 5^3$               | (48, 65) | 48  | $2 \cdot 3^2 \cdot 17^2$           |
| (16, 65) | 16  | $2^{16} \cdot 5^4$                      | (48, 66) | 48  | $2^4 \cdot 3^2 \cdot 17^2$         |
| (16, 68) | 16  | $2^{14} \cdot 5$                        | (48, 70) | 48  | $2^2 \cdot 3^2 \cdot 17$           |
| (16, 69) | 16  | $2^{14} \cdot 5^4 \cdot 11^2 \cdot 13$  | (48, 79) | 48  | $2^2 \cdot 3^2 \cdot 17$           |
| (16, 78) | 16  | $2^{12} \cdot 3^2 \cdot 5^3$            | (48, 80) | 48  | $2^3 \cdot 5 \cdot 13 \cdot 17$    |
| (16, 79) | 16  | $2^{16} \cdot 5^3$                      | (49, 51) | 49  | $2^4 \cdot 3^4 \cdot 5$            |
| (17, 18) | 17  | $2^6 \cdot 17$                          | (49, 58) | 49  | $2^3 \cdot 5 \cdot 13$             |
| (17, 29) | 17  | $2 \cdot 5^4$                           | (49, 69) | 49  | $2^3 \cdot 5 \cdot 13$             |
| (17, 30) | 17  | $2 \cdot 5^4$                           | (49, 74) | 49  | $3^2 \cdot 5 \cdot 17$             |
| (17, 45) | 17  | $3^2 \cdot 5^6$                         | (49, 78) | 49  | $29^2$                             |
| (17, 49) | 17  | $2^4 \cdot 3^4 \cdot 5$                 | (50, 55) | 50  | $2^2 \cdot 5^2$                    |
| (17, 50) | 17  | $3^4 \cdot 5^2$                         | (50, 56) | 50  | 17                                 |
| (17, 51) | 17  | $2^{16} \cdot 3^4$                      | (50, 66) | 50  | $2^5 \cdot 3^2 \cdot 5^2 \cdot 17$ |
| (17, 53) | 17  | $2^3 \cdot 3^2 \cdot 5^2$               | (50, 70) | 50  | $13 \cdot 17$                      |
| (17, 56) | 17  | $2^7 \cdot 3^6$                         | (50, 73) | 50  | $2^5 \cdot 41$                     |
| (17, 58) | 17  | $2^3 \cdot 5 \cdot 17$                  | (51, 58) | 51  | $2^3 \cdot 5 \cdot 17^2$           |
| (17, 59) | 17  | $2^3 \cdot 5 \cdot 17^2$                | (51, 59) | 51  | $2^3 \cdot 5 \cdot 17$             |
| (17, 64) | 17  | $2^3 \cdot 3^2 \cdot 5 \cdot 17$        | (51, 64) | 51  | $2^3 \cdot 3^2 \cdot 5^2$          |
| (17, 67) | 17  | $2^2 \cdot 3^4 \cdot 17$                | (51, 67) | 51  | $2^2 \cdot 3^4 \cdot 17$           |
| (17, 81) | 17  | $2^2 \cdot 3^4 \cdot 17$                | (51, 74) | 51  | $3^2 \cdot 17^2$                   |
| (18, 34) | 18  | $2^3 \cdot 3^2 \cdot 29$                | (51, 77) | 51  | $5 \cdot 17$                       |
| (18, 35) | 18  | $2^3 \cdot 5 \cdot 17 \cdot 37$         | (51, 79) | 51  | $2^2 \cdot 3^4 \cdot 5$            |
| (18, 46) | 18  | $2^3 \cdot 5^2$                         | (51, 81) | 51  | $2^2 \cdot 3^4 \cdot 17$           |
| (18, 47) | 18  | $2 \cdot 3^2 \cdot 5^2$                 | (52, 71) | 52  | $2^2 \cdot 17$                     |
| (18, 53) | 18  | $2 \cdot 5^2 \cdot 13 \cdot 17$         | (53, 54) | 53  | $2^2 \cdot 3^2 \cdot 17$           |
| (18, 58) | 18  | $2^4 \cdot 13$                          | (53, 60) | 53  | $2^3 \cdot 3^2 \cdot 13$           |
| (18, 64) | 18  | $2 \cdot 5 \cdot 17 \cdot 29$           | (53, 62) | 53  | $2^4 \cdot 3^4 \cdot 13$           |
| (18, 65) | 18  | $2^3 \cdot 5^3$                         | (53, 65) | 53  | $2 \cdot 3^2 \cdot 17$             |
| (18, 71) | 18  | 17                                      | (53, 71) | 53  | $17 \cdot 53$                      |
| (18, 75) | 18  | $13 \cdot 37$                           | (53, 78) | 53  | $3^2 \cdot 5^3$                    |
| (18, 76) | 18  | $2^4 \cdot 5 \cdot 17$                  | (53, 80) | 53  | $2^3 \cdot 3^4 \cdot 37$           |
| (18, 77) | 18  | $17^2$                                  | (54, 56) | 54  | $2^2 \cdot 3^2 \cdot 17$           |
| (19, 20) | 19  | $2^6$                                   | (54, 60) | 54  | $2^4 \cdot 3^4$                    |
| (19, 25) | 19  | $2^5 \cdot 7^2$                         | (54, 63) | 54  | $2^3 \cdot 3^2 \cdot 5^2$          |
| (19, 42) | 19  | $2^6 \cdot 5^3 \cdot 13^2$              | (54, 71) | 54  | $2^2 \cdot 17^2$                   |
| (19, 58) | 19  | $2^3 \cdot 13$                          | (55, 56) | 55  | $2^2 \cdot 17$                     |
| (19, 70) | 19  | $2^4 \cdot 5^4$                         | (55, 61) | 55  | $2^3 \cdot 13 \cdot 17$            |
| (19, 73) | 19  | 1                                       | (55, 73) | 55  | $2^4$                              |
| (20, 22) | 20  | $2^7 \cdot 5^3$                         | (55, 76) | 55  | $2^5 \cdot 5^2 \cdot 17^2$         |
| (20, 24) | 20  | $2^2 \cdot 3^2 \cdot 5^3 \cdot 17$      | (55, 77) | 55  | $2^3 \cdot 17$                     |

| $(i, j)$ | $k$ | $N(R_{ij})$                          | $(i, j)$ | $k$ | $N(R_{ij})$                       |
|----------|-----|--------------------------------------|----------|-----|-----------------------------------|
| (20, 25) | 20  | $2^2 \cdot 3^2 \cdot 5^3 \cdot 17$   | (55, 79) | 55  | $2^4 \cdot 5^5$                   |
| (20, 35) | 20  | $2^4 \cdot 5 \cdot 17^2$             | (56, 60) | 56  | $2^2 \cdot 13$                    |
| (20, 36) | 20  | $2^4 \cdot 5^2 \cdot 37^2$           | (56, 64) | 56  | $2^2 \cdot 11^2 \cdot 13^2$       |
| (20, 38) | 20  | $2^4 \cdot 5^6$                      | (56, 77) | 56  | $17^2$                            |
| (20, 39) | 20  | $5^2$                                | (57, 59) | 57  | $2^{10}$                          |
| (20, 43) | 20  | $2^6 \cdot 5^2$                      | (57, 62) | 57  | $2^{25}$                          |
| (20, 50) | 20  | $5^5$                                | (57, 63) | 57  | $2^3 \cdot 37 \cdot 61$           |
| (20, 51) | 20  | $2^7 \cdot 5^3$                      | (57, 64) | 57  | $2^3 \cdot 3^2 \cdot 5^2$         |
| (20, 56) | 20  | $2^{10}$                             | (57, 70) | 57  | $2^{10} \cdot 13^2$               |
| (20, 59) | 20  | $2^4 \cdot 3^2 \cdot 5^2$            | (57, 72) | 57  | $2^{11} \cdot 17 \cdot 37$        |
| (20, 63) | 20  | $2 \cdot 5$                          | (57, 81) | 57  | $2^6 \cdot 3^4$                   |
| (20, 65) | 20  | $2 \cdot 37$                         | (58, 65) | 58  | $2^6 \cdot 17$                    |
| (20, 66) | 20  | $2^7 \cdot 37$                       | (58, 67) | 58  | $2^7 \cdot 17^2$                  |
| (20, 68) | 20  | $2 \cdot 17$                         | (58, 68) | 58  | $2^5$                             |
| (21, 23) | 21  | $2^2 \cdot 3^4 \cdot 5^2$            | (58, 74) | 58  | $5 \cdot 17$                      |
| (21, 26) | 21  | $2^2 \cdot 3^2 \cdot 5^2 \cdot 17^2$ | (59, 66) | 59  | $2^{11} \cdot 17$                 |
| (21, 28) | 21  | $2 \cdot 5 \cdot 17$                 | (59, 77) | 59  | $5 \cdot 17$                      |
| (21, 32) | 21  | $2 \cdot 3^2 \cdot 29 \cdot 37$      | (60, 64) | 60  | $2^3 \cdot 3^2 \cdot 13$          |
| (21, 39) | 21  | $3^4 \cdot 5^8$                      | (60, 77) | 60  | $2^2 \cdot 3^2$                   |
| (21, 42) | 21  | $2^6 \cdot 5^2 \cdot 29$             | (61, 62) | 61  | $2^6 \cdot 5^6$                   |
| (21, 45) | 21  | $3^4 \cdot 5^4$                      | (61, 64) | 61  | $2 \cdot 3^4 \cdot 13$            |
| (21, 47) | 21  | $2 \cdot 5 \cdot 17$                 | (62, 67) | 62  | $2^{16} \cdot 3^2$                |
| (21, 53) | 21  | $2^2 \cdot 5^4$                      | (62, 68) | 62  | $2^9 \cdot 17 \cdot 41$           |
| (21, 58) | 21  | $2^4$                                | (62, 70) | 62  | $2^{12} \cdot 5^2 \cdot 17^2$     |
| (21, 60) | 21  | $2^2 \cdot 3^6$                      | (62, 72) | 62  | $2^{13} \cdot 13$                 |
| (21, 64) | 21  | $2^2 \cdot 5^4$                      | (62, 76) | 62  | $2^{18} \cdot 17$                 |
| (21, 65) | 21  | $2 \cdot 3^2 \cdot 5^2$              | (63, 72) | 63  | $2^5 \cdot 5^3 \cdot 37$          |
| (21, 66) | 21  | $2^7 \cdot 3^2 \cdot 5^2$            | (64, 65) | 64  | $2 \cdot 5^2 \cdot 29$            |
| (21, 69) | 21  | $2^2 \cdot 5^2 \cdot 13 \cdot 17$    | (64, 66) | 64  | $2^4 \cdot 3^2 \cdot 17$          |
| (21, 77) | 21  | $3^2$                                | (64, 72) | 64  | $2^2 \cdot 17 \cdot 29$           |
| (21, 80) | 21  | $2^6 \cdot 5 \cdot 13 \cdot 37$      | (64, 78) | 64  | $29 \cdot 41$                     |
| (22, 30) | 22  | $2^{20} \cdot 5$                     | (65, 67) | 65  | $2^8 \cdot 17^2$                  |
| (22, 35) | 22  | $2^{21} \cdot 5$                     | (65, 72) | 65  | $2^4 \cdot 5^2 \cdot 17 \cdot 29$ |
| (22, 39) | 22  | $2^{18} \cdot 5 \cdot 37 \cdot 41$   | (65, 76) | 65  | $2^{11} \cdot 5$                  |
| (22, 68) | 22  | $2^{23} \cdot 41$                    | (66, 73) | 66  | $2^8 \cdot 17$                    |
| (22, 79) | 22  | $2^{20} \cdot 5^4$                   | (66, 74) | 66  | $2^{11} \cdot 17^2$               |
| (22, 80) | 22  | $2^{24} \cdot 3^4 \cdot 37$          | (66, 81) | 66  | $2^{16} \cdot 17^2$               |
| (23, 29) | 23  | $2^8 \cdot 5^4$                      | (67, 77) | 67  | $2^3 \cdot 17^2$                  |
| (23, 30) | 23  | $2^8 \cdot 5^2$                      | (69, 70) | 69  | $2^2 \cdot 5^2$                   |
| (23, 37) | 23  | $2^9 \cdot 5^3$                      | (71, 77) | 71  | $17 \cdot 53$                     |
| (23, 39) | 23  | $2^8 \cdot 3^4$                      | (73, 77) | 73  | $17 \cdot 53$                     |
| (23, 44) | 23  | $2^4$                                | (74, 76) | 74  | $2^2 \cdot 5^2$                   |
| (23, 45) | 23  | $2^4 \cdot 3^4 \cdot 5^4$            | (74, 77) | 74  | $2^5 \cdot 5 \cdot 17$            |
| (23, 49) | 23  | $2^2 \cdot 5^2$                      | (77, 81) | 77  | $2^3 \cdot 17^2$                  |

Table 4: The terminal residue classes  $x \equiv a, y \equiv b \pmod{p^k}$  of Step 4, grouped by  $p, k$  and the value of  $B_p$ .

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                   |
|-----|-----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2   | 6   | 1     | (0, 43), (1, 43), (4, 55), (5, 27), (8, 35), (9, 11), (12, 47), (13, 59),<br>(16, 27), (17, 43), (20, 39), (21, 27), (24, 19), (25, 11), (28, 31),<br>(29, 59), (32, 11), (33, 43), (36, 23), (37, 27), (40, 3), (41, 11), (44, 15),<br>(45, 59), (48, 59), (49, 43), (52, 7), (53, 27), (56, 51), (57, 11), (60, 63),<br>(61, 59) |

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|-----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2   | 6   | 2     | (2, 53), (3, 27), (6, 33), (7, 11), (10, 45), (11, 59), (14, 25), (15, 43),<br>(18, 37), (19, 27), (22, 17), (23, 11), (26, 29), (27, 59), (30, 9), (31, 43),<br>(34, 21), (35, 27), (38, 1), (39, 11), (42, 13), (43, 59), (46, 57), (47, 43),<br>(50, 5), (51, 27), (54, 49), (55, 11), (58, 61), (59, 59), (62, 41), (63, 43)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 3   | 2   | 2     | (0, 4), (3, 7)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3   | 2   | 3     | (1, 4), (2, 1), (4, 1), (5, 7), (7, 7), (8, 4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 3   | 3   | 2     | (6, 1), (15, 10), (24, 19)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 5   | 3   | 2     | (0, 13), (3, 23), (4, 93), (6, 93), (7, 88), (9, 98), (10, 23), (11, 98),<br>(12, 43), (15, 103), (16, 28), (17, 48), (18, 88), (19, 8), (20, 108),<br>(21, 8), (22, 103), (23, 43), (24, 38), (25, 38), (26, 38), (27, 83),<br>(29, 118), (31, 118), (32, 113), (33, 103), (34, 123), (35, 48), (37, 68),<br>(40, 3), (41, 53), (42, 73), (43, 113), (44, 33), (46, 33), (47, 3), (49, 63),<br>(50, 63), (51, 63), (52, 108), (54, 18), (56, 18), (57, 13), (59, 23),<br>(60, 73), (61, 23), (65, 28), (66, 78), (67, 98), (68, 13), (71, 58),<br>(72, 28), (74, 88), (75, 88), (76, 88), (77, 8), (78, 98), (79, 43), (81, 43),<br>(82, 38), (84, 48), (85, 98), (86, 48), (87, 118), (88, 8), (90, 53),<br>(91, 103), (92, 123), (93, 38), (94, 83), (96, 83), (97, 53), (99, 113),<br>(100, 113), (101, 113), (102, 33), (104, 68), (106, 68), (107, 63),<br>(109, 73), (110, 123), (111, 73), (112, 18), (115, 78), (116, 3), (117, 23),<br>(118, 63), (119, 108), (121, 108), (122, 78), (124, 13)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 5   | 4   | 2     | (2, 308), (8, 578), (13, 308), (14, 153), (28, 173), (36, 123), (38, 333),<br>(39, 553), (45, 133), (48, 193), (53, 323), (58, 128), (62, 93), (63, 358),<br>(64, 328), (69, 308), (70, 283), (73, 93), (83, 528), (89, 103), (95, 433),<br>(98, 618), (103, 623), (108, 303), (113, 408), (120, 583), (123, 518),<br>(126, 388), (127, 433), (133, 78), (138, 433), (139, 278), (153, 298),<br>(161, 248), (163, 458), (164, 53), (170, 258), (173, 318), (178, 448),<br>(183, 253), (187, 218), (188, 483), (189, 453), (194, 433), (195, 408),<br>(198, 218), (208, 28), (214, 228), (220, 558), (223, 118), (228, 123),<br>(233, 428), (238, 533), (245, 83), (248, 18), (251, 513), (252, 558),<br>(258, 203), (263, 558), (278, 423), (286, 373), (288, 583), (289, 178),<br>(295, 383), (298, 443), (303, 573), (308, 378), (312, 343), (313, 608),<br>(314, 578), (319, 558), (320, 533), (323, 343), (333, 153), (339, 353),<br>(345, 58), (348, 243), (353, 248), (358, 553), (363, 33), (370, 208),<br>(373, 143), (376, 13), (377, 58), (383, 328), (388, 58), (389, 528),<br>(403, 548), (411, 498), (413, 83), (414, 303), (420, 508), (423, 568),<br>(428, 73), (433, 503), (437, 468), (438, 108), (439, 78), (444, 58),<br>(445, 33), (448, 468), (458, 278), (464, 478), (470, 183), (473, 368),<br>(478, 373), (483, 53), (488, 158), (495, 333), (498, 268), (501, 138),<br>(502, 183), (508, 453), (513, 183), (514, 28), (528, 48), (536, 623),<br>(538, 208), (539, 428), (545, 8), (548, 68), (553, 198), (558, 3),<br>(562, 593), (563, 233), (564, 203), (569, 183), (570, 158), (573, 593),<br>(583, 403), (589, 603), (595, 308), (598, 493), (603, 498), (608, 178),<br>(613, 283), (620, 458), (623, 393) |

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5   | 5   | 2     | (1, 1513), (5, 368), (30, 393), (55, 1668), (105, 1718), (130, 493),<br>(155, 518), (180, 1793), (230, 1843), (239, 628), (255, 618),<br>(264, 2903), (280, 643), (305, 1918), (355, 1968), (364, 2628),<br>(380, 743), (405, 768), (430, 2043), (480, 2093), (489, 1503),<br>(505, 868), (530, 893), (555, 2168), (605, 2218), (614, 378),<br>(626, 2138), (630, 993), (655, 1018), (680, 2293), (730, 2343),<br>(739, 2378), (755, 1118), (780, 1143), (805, 2418), (855, 2468),<br>(864, 1253), (880, 1243), (889, 403), (905, 1268), (930, 2543),<br>(980, 2593), (989, 128), (1005, 1368), (1030, 1393), (1055, 2668),<br>(1105, 2718), (1114, 2128), (1130, 1493), (1155, 1518), (1180, 2793),<br>(1230, 2843), (1239, 1003), (1251, 2763), (1255, 1618), (1280, 1643),<br>(1305, 2918), (1355, 2968), (1364, 3003), (1380, 1743), (1405, 1768),<br>(1430, 3043), (1480, 3093), (1489, 1878), (1505, 1868), (1530, 1893),<br>(1555, 43), (1605, 93), (1614, 753), (1630, 1993), (1655, 2018),<br>(1680, 168), (1730, 218), (1739, 2753), (1755, 2118), (1780, 2143),<br>(1805, 293), (1855, 343), (1864, 1628), (1876, 263), (1880, 2243),<br>(1905, 2268), (1930, 418), (1980, 468), (2005, 2368), (2030, 2393),<br>(2055, 543), (2105, 593), (2114, 2503), (2130, 2493), (2139, 1653),<br>(2155, 2518), (2180, 668), (2230, 718), (2239, 1378), (2255, 2618),<br>(2280, 2643), (2305, 793), (2355, 843), (2364, 253), (2380, 2743),<br>(2405, 2768), (2430, 918), (2480, 968), (2489, 2253), (2501, 888),<br>(2505, 2868), (2530, 2893), (2555, 1043), (2605, 1093), (2630, 2993),<br>(2655, 3018), (2680, 1168), (2730, 1218), (2739, 3), (2755, 3118),<br>(2764, 2278), (2780, 18), (2805, 1293), (2855, 1343), (2864, 2003),<br>(2880, 118), (2905, 143), (2930, 1418), (2980, 1468), (2989, 878),<br>(3005, 243), (3030, 268), (3055, 1543), (3105, 1593), (3114, 2878) |

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|-----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5   | 6   | 2     | (80, 10443), (114, 4878), (205, 10568), (330, 10693), (455, 10818),<br>(580, 10943), (705, 11068), (830, 11193), (955, 11318), (1080, 11443),<br>(1205, 11568), (1330, 11693), (1455, 11818), (1514, 13528),<br>(1580, 11943), (1705, 12068), (1830, 12193), (1955, 12318),<br>(1989, 3628), (2080, 12443), (2205, 12568), (2330, 12693),<br>(2455, 12818), (2580, 12943), (2614, 13628), (2705, 13068),<br>(2830, 13193), (2955, 13318), (3080, 13443), (3205, 13568),<br>(3239, 8003), (3330, 13693), (3455, 13818), (3580, 13943),<br>(3705, 14068), (3830, 14193), (3955, 14318), (4080, 14443),<br>(4205, 14568), (4330, 14693), (4455, 14818), (4580, 14943),<br>(4639, 1028), (4705, 15068), (4830, 15193), (4955, 15318),<br>(5080, 15443), (5114, 6753), (5205, 15568), (5330, 68), (5455, 193),<br>(5580, 318), (5705, 443), (5739, 1128), (5830, 568), (5955, 693),<br>(6080, 818), (6205, 943), (6330, 1068), (6364, 11128), (6455, 1193),<br>(6580, 1318), (6705, 1443), (6830, 1568), (6955, 1693), (7080, 1818),<br>(7205, 1943), (7330, 2068), (7455, 2193), (7580, 2318), (7705, 2443),<br>(7764, 4153), (7830, 2568), (7955, 2693), (8080, 2818), (8205, 2943),<br>(8239, 9878), (8330, 3068), (8455, 3193), (8580, 3318), (8705, 3443),<br>(8830, 3568), (8864, 4253), (8955, 3693), (9080, 3818), (9205, 3943),<br>(9330, 4068), (9455, 4193), (9489, 14253), (9580, 4318), (9705, 4443),<br>(9830, 4568), (9955, 4693), (10080, 4818), (10205, 4943),<br>(10330, 5068), (10455, 5193), (10580, 5318), (10705, 5443),<br>(10830, 5568), (10889, 7278), (10955, 5693), (11080, 5818),<br>(11205, 5943), (11330, 6068), (11364, 13003), (11455, 6193),<br>(11580, 6318), (11705, 6443), (11830, 6568), (11955, 6693),<br>(11989, 7378), (12080, 6818), (12205, 6943), (12330, 7068),<br>(12455, 7193), (12580, 7318), (12614, 1753), (12705, 7443),<br>(12830, 7568), (12955, 7693), (13080, 7818), (13205, 7943),<br>(13330, 8068), (13455, 8193), (13580, 8318), (13705, 8443),<br>(13830, 8568), (13955, 8693), (14014, 10403), (14080, 8818),<br>(14205, 8943), (14330, 9068), (14455, 9193), (14489, 503),<br>(14580, 9318), (14705, 9443), (14830, 9568), (14955, 9693),<br>(15080, 9818), (15114, 10503), (15205, 9943), (15330, 10068),<br>(15455, 10193), (15580, 10318) |
| 7   | 1   | 0     | (2, 3), (3, 3)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 11  | 1   | 0     | (2, 5), (4, 5), (5, 4), (6, 3), (6, 9), (6, 10), (8, 7)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 13  | 2   | 0     | (2, 116), (3, 136), (4, 123), (6, 54), (7, 38), (9, 40), (10, 105), (11, 2),<br>(15, 90), (16, 71), (17, 136), (19, 145), (20, 77), (22, 79), (23, 79),<br>(24, 54), (28, 64), (29, 6), (30, 149), (32, 67), (33, 116), (35, 118),<br>(36, 53), (37, 106), (41, 38), (42, 110), (43, 162), (45, 158), (46, 155),<br>(48, 157), (49, 27), (50, 158), (54, 12), (55, 45), (56, 6), (58, 80),<br>(59, 25), (61, 27), (63, 41), (67, 155), (68, 149), (69, 19), (72, 64),<br>(74, 66), (75, 144), (76, 93), (80, 129), (81, 84), (82, 32), (84, 93),<br>(85, 103), (87, 105), (88, 118), (89, 145), (93, 103), (94, 19), (95, 45),<br>(97, 15), (98, 142), (100, 144), (101, 92), (102, 28), (106, 77),<br>(107, 123), (108, 58), (110, 106), (111, 12), (113, 14), (114, 66),<br>(115, 80), (119, 51), (120, 58), (121, 71), (124, 51), (126, 53),<br>(128, 132), (132, 25), (133, 162), (134, 84), (137, 90), (139, 92),<br>(140, 14), (141, 15), (146, 97), (147, 97), (149, 41), (150, 129),<br>(152, 131), (153, 157), (154, 67), (158, 142), (159, 32), (160, 110),<br>(162, 132), (163, 168), (165, 1), (166, 131)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----|-----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13  | 3   | 0     | (62, 1691), (71, 678), (123, 704), (127, 1899), (136, 457), (145, 1013),<br>(167, 626), (231, 1353), (240, 1861), (292, 1887), (296, 1561),<br>(305, 1640), (314, 675), (336, 1302), (400, 1015), (409, 847), (461, 873),<br>(465, 1223), (474, 626), (483, 337), (505, 1978), (569, 677), (578, 2030),<br>(630, 2056), (634, 885), (643, 1809), (652, 2196), (674, 457), (738, 339),<br>(747, 1016), (799, 1042), (803, 547), (812, 795), (821, 1858),<br>(843, 1133), (916, 2), (968, 28), (972, 209), (981, 1978), (990, 1520),<br>(1012, 1809), (1076, 1860), (1085, 1185), (1137, 1211), (1141, 2068),<br>(1150, 964), (1159, 1182), (1181, 288), (1245, 1522), (1254, 171),<br>(1306, 197), (1310, 1730), (1319, 2147), (1328, 844), (1350, 964),<br>(1414, 1184), (1423, 1354), (1475, 1380), (1479, 1392), (1488, 1133),<br>(1497, 506), (1519, 1640), (1583, 846), (1592, 340), (1644, 366),<br>(1648, 1054), (1657, 119), (1666, 168), (1688, 119), (1752, 508),<br>(1761, 1523), (1813, 1549), (1817, 716), (1826, 1302), (1835, 2027),<br>(1857, 795), (1921, 170), (1930, 509), (1982, 535), (1986, 378),<br>(1995, 288), (2004, 1689), (2026, 1471), (2090, 2029), (2099, 1692),<br>(2151, 1718), (2155, 40), (2164, 1471), (2173, 1351), (2195, 2147)<br>(907, 8789), (3104, 4395), (5301, 1), (7498, 24168), (9695, 19774),<br>(11892, 15380), (14089, 10986), (16286, 6592), (18483, 2198),<br>(20680, 26365), (22877, 21971), (25074, 17577), (27271, 13183) |
| 17  | 1   | 0     | (0, 2), (1, 2), (4, 2), (7, 4), (13, 2), (15, 4)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17  | 2   | 0     | (3, 24), (6, 79), (7, 118), (7, 286), (8, 181), (9, 43), (9, 51), (9, 195),<br>(10, 95), (11, 77), (11, 229), (11, 272), (12, 114), (14, 253), (15, 101),<br>(15, 133), (16, 87), (19, 61), (20, 279), (23, 249), (24, 201), (24, 237),<br>(25, 249), (26, 110), (26, 221), (26, 247), (27, 112), (28, 9), (28, 119),<br>(28, 161), (29, 199), (31, 185), (32, 33), (33, 155), (36, 10), (37, 245),<br>(40, 130), (41, 67), (42, 28), (43, 25), (43, 102), (43, 162), (44, 129),<br>(45, 93), (45, 230), (45, 255), (46, 284), (48, 117), (49, 65), (49, 254),<br>(50, 223), (53, 248), (54, 211), (57, 11), (58, 31), (58, 186), (59, 96),<br>(60, 77), (60, 229), (60, 272), (61, 146), (62, 25), (62, 102), (62, 162),<br>(63, 80), (65, 49), (66, 31), (66, 186), (67, 2), (70, 197), (71, 177),<br>(74, 181), (75, 16), (75, 235), (76, 164), (77, 144), (77, 153), (77, 281),<br>(78, 163), (79, 94), (79, 238), (79, 246), (80, 165), (82, 270), (83, 118),<br>(83, 286), (84, 70), (88, 143), (91, 62), (92, 135), (92, 150), (93, 232),<br>(94, 34), (94, 59), (94, 196), (95, 180), (96, 85), (96, 178), (97, 250),<br>(99, 202), (100, 50), (100, 252), (101, 138), (104, 95), (105, 109),<br>(108, 232), (109, 65), (109, 254), (110, 11), (111, 204), (111, 263),<br>(112, 197), (113, 110), (113, 221), (113, 247), (114, 46), (116, 134),<br>(117, 218), (117, 271), (118, 206), (121, 44), (122, 75), (125, 113),<br>(126, 84), (127, 79), (128, 26), (128, 85), (128, 178), (129, 214),<br>(130, 42), (130, 68), (130, 179), (131, 131), (133, 66), (134, 184),<br>(134, 203), (135, 274), (138, 282), (139, 41), (142, 283), (143, 184),<br>(143, 203), (144, 147), (145, 93), (145, 230), (145, 255), (147, 111),<br>(147, 204), (147, 263), (148, 216), (151, 135), (151, 150), (152, 53),<br>(155, 231), (156, 7), (159, 164), (160, 33), (160, 99), (161, 215),<br>(162, 8), (162, 136), (162, 145), (163, 248), (164, 43), (164, 51),<br>(164, 195), (165, 12), (167, 219), (168, 67), (168, 116), (169, 121),<br>(172, 180), (173, 262), (176, 45), (177, 14), (177, 152), (178, 283),<br>(179, 17), (179, 60), (179, 212), (180, 265), (181, 127), (181, 187),<br>(181, 264), (182, 97), (184, 151), (185, 82), (185, 288), (186, 189),<br>(189, 129), (190, 228), (193, 215), (194, 218), (194, 271), (195, 62),<br>(196, 127), (196, 187), (196, 264), (197, 282), (198, 34), (198, 59),<br>(198, 196), (199, 182), (201, 83), (202, 48), (202, 220), (203, 257),<br>(206, 78), (207, 194), (210, 96), (211, 101), (211, 133), (212, 130),<br>(213, 68), (213, 179), (214, 10), (215, 128), (215, 170), (215, 280),<br>(216, 267), (218, 15), (219, 14), (219, 152), (220, 36), (223, 27),<br>(224, 160), (227, 266), (228, 48), (228, 220), (229, 198), (230, 94),<br>(230, 238), (230, 246), (231, 27), (232, 17), (232, 60), (232, 212),<br>(233, 63), (235, 236), (236, 84), (236, 269), (237, 104), (240, 265),<br>(241, 126), (244, 147), (245, 50), (245, 252), (246, 266), (247, 9),<br>(247, 119), (247, 161), (248, 44), (249, 144), (249, 153), (249, 281),<br>(250, 148), (252, 168), (253, 16), (253, 235), (254, 172), (257, 214),<br>(258, 92), (261, 28), (262, 167), (262, 169), (263, 45), (264, 0),<br>(264, 76), (265, 61), (266, 0), (266, 76), (266, 213), (267, 233),<br>(269, 100), (270, 201), (270, 237), (271, 240), (274, 163), (275, 58),<br>(278, 198), (279, 82), (279, 288), (280, 113), (281, 128), (281, 170),<br>(281, 280), (282, 78), (283, 8), (283, 136), (283, 145), (284, 29),<br>(286, 32), (287, 167), (287, 169), (288, 19) |



| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17  | 3   | 0     | (2, 2713), (32, 3856), (41, 694), (87, 1591), (96, 893), (111, 3290),<br>(126, 4893), (146, 1676), (150, 287), (213, 1776), (264, 4548),<br>(291, 1846), (321, 3278), (330, 4162), (376, 724), (385, 4650),<br>(400, 1845), (415, 3448), (435, 1965), (439, 4044), (502, 331),<br>(553, 3103), (580, 979), (610, 2700), (619, 2717), (665, 4770),<br>(674, 3494), (689, 400), (704, 2003), (724, 2254), (728, 2888),<br>(791, 3799), (842, 1658), (869, 112), (899, 2122), (908, 1272),<br>(954, 3903), (963, 2338), (978, 3868), (993, 558), (1013, 2543),<br>(1017, 1732), (1080, 2354), (1131, 213), (1158, 4158), (1188, 1544),<br>(1197, 4740), (1243, 3036), (1252, 1182), (1267, 2423), (1282, 4026),<br>(1302, 2832), (1306, 576), (1369, 909), (1420, 3681), (1447, 3291),<br>(1477, 966), (1486, 3295), (1532, 2169), (1541, 26), (1556, 978),<br>(1571, 2581), (1591, 3121), (1595, 4333), (1658, 4377), (1709, 2236),<br>(1736, 2424), (1766, 388), (1775, 1850), (1821, 1302), (1830, 3783),<br>(1845, 4446), (1860, 1136), (1880, 3410), (1884, 3177), (1947, 2932),<br>(1998, 791), (2025, 1557), (2055, 4723), (2064, 405), (2110, 435),<br>(2119, 2627), (2134, 3001), (2149, 4604), (2169, 3699), (2173, 2021),<br>(2236, 1487), (2287, 4259), (2314, 690), (2344, 4145), (2353, 3873),<br>(2399, 4481), (2408, 1471), (2423, 1556), (2438, 3159), (2458, 3988),<br>(2462, 865), (2525, 42), (2576, 2814), (2603, 4736), (2633, 3567),<br>(2642, 2428), (2688, 3614), (2697, 315), (2712, 111), (2727, 1714),<br>(2747, 4277), (2751, 4622), (2814, 3510), (2865, 1369), (2892, 3869),<br>(2922, 2989), (2931, 983), (2977, 2747), (2986, 4072), (3001, 3579),<br>(3016, 269), (3036, 4566), (3040, 3466), (3103, 2065), (3154, 4837),<br>(3181, 3002), (3211, 2411), (3220, 4451), (3266, 1880), (3275, 2916),<br>(3290, 2134), (3305, 3737), (3325, 4855), (3329, 2310), (3392, 620),<br>(3443, 3392), (3470, 2135), (3500, 1833), (3509, 3006), (3555, 1013),<br>(3564, 1760), (3579, 689), (3594, 2292), (3614, 231), (3618, 1154),<br>(3681, 4088), (3732, 1947), (3759, 1268), (3789, 1255), (3798, 1561),<br>(3844, 146), (3853, 604), (3868, 4157), (3883, 847), (3903, 520),<br>(3907, 4911), (3970, 2643), (4021, 502), (4048, 401), (4078, 677),<br>(4087, 116), (4133, 4192), (4142, 4361), (4157, 2712), (4172, 4315),<br>(4192, 809), (4196, 3755), (4259, 1198), (4310, 3970), (4337, 4447),<br>(4367, 99), (4376, 3584), (4422, 3325), (4431, 3205), (4446, 1267),<br>(4461, 2870), (4481, 1098), (4485, 2599), (4548, 4666), (4599, 2525),<br>(4626, 3580), (4656, 4434), (4665, 2139), (4711, 2458), (4720, 2049),<br>(4735, 4735), (4750, 1425), (4770, 1387), (4774, 1443), (4837, 3221),<br>(4888, 1080) |
| 29  | 1   | 0     | (4, 7), (7, 26), (11, 10), (13, 22), (14, 7), (15, 19), (16, 23), (21, 20),<br>(22, 6), (24, 14), (25, 19), (26, 12), (26, 21), (26, 25), (27, 0)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29  | 2   | 0     | (5, 647), (20, 724), (24, 263), (24, 448), (27, 411), (27, 575), (34, 270),<br>(49, 463), (53, 274), (53, 698), (56, 382), (56, 546), (63, 734), (78, 202),<br>(82, 100), (82, 292), (85, 353), (85, 517), (92, 357), (107, 782),<br>(111, 727), (111, 767), (114, 324), (114, 488), (121, 821), (136, 521),<br>(140, 321), (140, 593), (143, 295), (143, 459), (150, 444), (165, 260),<br>(169, 419), (169, 756), (172, 266), (172, 430), (179, 67), (194, 840),<br>(198, 245), (198, 350), (201, 237), (201, 401), (208, 531), (223, 579),<br>(227, 71), (230, 208), (230, 372), (237, 154), (252, 318), (256, 379),<br>(256, 738), (259, 179), (259, 343), (266, 618), (281, 57), (285, 564),<br>(285, 814), (288, 150), (288, 314), (295, 241), (310, 637), (314, 390),<br>(314, 408), (317, 121), (317, 285), (324, 705), (339, 376), (343, 2),<br>(343, 216), (346, 92), (346, 256), (353, 328), (368, 115), (372, 42),<br>(372, 437), (375, 63), (375, 227), (382, 792), (397, 695), (401, 31),<br>(401, 709), (404, 34), (404, 198), (411, 415), (426, 434), (430, 466),<br>(430, 535), (433, 5), (433, 169), (440, 38), (455, 173), (459, 60),<br>(459, 361), (462, 140), (462, 817), (469, 502), (484, 753), (488, 187),<br>(488, 495), (491, 111), (491, 788), (498, 125), (513, 492), (517, 13),<br>(517, 89), (520, 82), (520, 759), (527, 589), (542, 231), (546, 524),<br>(546, 680), (549, 53), (549, 730), (556, 212), (571, 811), (575, 118),<br>(575, 506), (578, 24), (578, 701), (585, 676), (600, 550), (604, 332),<br>(604, 553), (607, 672), (607, 836), (614, 299), (629, 289), (633, 147),<br>(633, 158), (636, 643), (636, 807), (643, 763), (658, 28), (662, 582),<br>(662, 825), (665, 614), (665, 778), (672, 386), (687, 608), (691, 176),<br>(691, 651), (694, 585), (694, 749), (701, 9), (716, 347), (720, 477),<br>(720, 611), (723, 556), (723, 720), (730, 473), (745, 86), (749, 205),<br>(749, 303), (752, 527), (752, 691), (759, 96), (774, 666), (778, 129),<br>(778, 640), (781, 498), (781, 662), (788, 560), (803, 405), (807, 234),<br>(807, 796), (810, 469), (810, 633), (817, 183), (832, 144), (836, 622),<br>(836, 669), (839, 440), (839, 604) |
| 29  | 3   | 0     | (227, 20128), (1068, 8354), (1909, 20969), (2750, 9195), (3591, 21810),<br>(4432, 10036), (5273, 22651), (6114, 10877), (6955, 23492),<br>(7796, 11718), (8637, 24333), (9478, 12559), (10319, 785),<br>(11160, 13400), (12001, 1626), (12842, 14241), (13683, 2467),<br>(14524, 15082), (15365, 3308), (16206, 15923), (17047, 4149),<br>(17888, 16764), (18729, 4990), (19570, 17605), (20411, 5831),<br>(21252, 18446), (22093, 6672), (22934, 19287), (23775, 7513)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 37  | 1   | 0     | (0, 30), (1, 30), (2, 4), (2, 11), (2, 22), (3, 6), (6, 30), (7, 5), (8, 18),<br>(8, 35), (13, 20), (14, 18), (14, 21), (14, 35), (16, 31), (17, 5), (18, 3),<br>(20, 7), (21, 2), (21, 16), (21, 19), (22, 24), (23, 20), (24, 18), (24, 21),<br>(25, 26), (25, 33), (29, 20), (30, 7), (31, 30), (32, 29), (35, 13), (36, 30)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 37  | 2   | 0     | (8, 502), (9, 399), (12, 1120), (19, 175), (24, 738), (25, 237), (45, 909),<br>(46, 325), (49, 1268), (56, 360), (61, 775), (62, 644), (82, 1316),<br>(83, 251), (86, 47), (93, 545), (98, 812), (99, 1051), (119, 354),<br>(120, 177), (123, 195), (130, 730), (135, 849), (136, 89), (156, 761),<br>(157, 103), (160, 343), (167, 915), (172, 886), (173, 496), (193, 1168),<br>(194, 29), (197, 491), (204, 1100), (209, 923), (210, 903), (230, 206),<br>(231, 1324), (234, 639), (241, 1285), (246, 960), (247, 1310),<br>(267, 613), (268, 1250), (271, 787), (278, 101), (283, 997), (284, 348),<br>(304, 1020), (305, 1176), (308, 935), (315, 286), (320, 1034),<br>(321, 755), (341, 58), (342, 1102), (345, 1083), (352, 471), (357, 1071),<br>(358, 1162), (378, 465), (379, 1028), (382, 1231), (389, 656),<br>(394, 1108), (395, 200), (415, 872), (416, 954), (419, 10), (426, 841),<br>(431, 1145), (432, 607), (452, 1279), (453, 880), (456, 158),<br>(463, 1026), (468, 1182), (469, 1014), (489, 317), (490, 806),<br>(493, 306), (500, 1211), (505, 1219), (506, 52), (526, 724), (527, 732),<br>(530, 454), (537, 27), (542, 1256), (543, 459), (563, 1131), (564, 658),<br>(567, 602), (574, 212), (579, 1293), (580, 866), (600, 169), (601, 584),<br>(604, 750), (611, 397), (616, 1330), (617, 1273), (637, 576), (638, 510),<br>(641, 898), (648, 582), (653, 1367), (654, 311), (674, 983), (675, 436),<br>(678, 1046), (685, 767), (690, 35), (691, 718), (711, 21), (712, 362),<br>(715, 1194), (722, 952), (727, 72), (728, 1125), (748, 428), (749, 288),<br>(752, 1342), (759, 1137), (764, 109), (765, 163), (785, 835), (786, 214),<br>(789, 121), (796, 1322), (801, 146), (802, 570), (822, 1242), (823, 140),<br>(826, 269), (833, 138), (838, 183), (839, 977), (859, 280), (860, 66),<br>(863, 417), (870, 323), (875, 220), (876, 15), (896, 687), (897, 1361),<br>(900, 565), (907, 508), (912, 257), (913, 422), (933, 1094), (934, 1287),<br>(937, 713), (944, 693), (949, 294), (950, 829), (970, 132), (971, 1213),<br>(974, 861), (981, 878), (986, 331), (987, 1236), (1007, 539),<br>(1008, 1139), (1011, 1009), (1018, 1063), (1023, 368), (1024, 274),<br>(1044, 946), (1045, 1065), (1048, 1157), (1055, 1248), (1060, 405),<br>(1061, 681), (1081, 1353), (1082, 991), (1085, 1305), (1092, 64),<br>(1097, 442), (1098, 1088), (1118, 391), (1119, 917), (1122, 84),<br>(1129, 249), (1134, 479), (1135, 126), (1155, 798), (1156, 843),<br>(1159, 232), (1166, 434), (1171, 516), (1172, 533), (1192, 1205),<br>(1193, 769), (1196, 380), (1203, 619), (1208, 553), (1209, 940),<br>(1229, 243), (1230, 695), (1233, 528), (1240, 804), (1245, 590),<br>(1246, 1347), (1266, 650), (1267, 621), (1270, 676), (1277, 989),<br>(1282, 627), (1283, 385), (1303, 1057), (1304, 547), (1307, 824),<br>(1314, 1174), (1319, 664), (1320, 792), (1340, 95), (1341, 473),<br>(1344, 972), (1351, 1359), (1356, 701), (1357, 1199) |
| 41  | 1   | 0     | (5, 40), (6, 16), (6, 30), (6, 36), (7, 26), (8, 3), (8, 4), (13, 10), (14, 9),<br>(15, 0), (15, 18), (15, 23), (16, 32), (19, 15), (20, 29), (21, 8), (25, 39),<br>(26, 3), (26, 4), (26, 34), (27, 14), (30, 15), (33, 0), (33, 18), (35, 17),<br>(36, 20), (37, 6), (37, 13), (37, 22), (39, 27)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|-----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 41  | 2   | 0     | (3, 206), (8, 157), (17, 238), (18, 1150), (24, 805), (33, 925), (34, 1386),<br>(44, 1), (49, 608), (58, 1632), (59, 2), (65, 1502), (74, 187), (75, 1345),<br>(85, 1477), (90, 1059), (99, 1345), (100, 535), (106, 518), (115, 1130),<br>(116, 1304), (126, 1272), (131, 1510), (140, 1058), (141, 1068),<br>(147, 1215), (156, 392), (157, 1263), (167, 1067), (172, 280), (181, 771),<br>(182, 1601), (188, 231), (197, 1335), (198, 1222), (208, 862),<br>(213, 731), (222, 484), (223, 453), (229, 928), (238, 597), (239, 1181),<br>(249, 657), (254, 1182), (263, 197), (264, 986), (270, 1625),<br>(279, 1540), (280, 1140), (290, 452), (295, 1633), (304, 1591),<br>(305, 1519), (311, 641), (320, 802), (321, 1099), (331, 247), (336, 403),<br>(345, 1304), (346, 371), (352, 1338), (361, 64), (362, 1058), (372, 42),<br>(377, 854), (386, 1017), (387, 904), (393, 354), (402, 1007),<br>(403, 1017), (413, 1518), (418, 1305), (427, 730), (428, 1437),<br>(434, 1051), (443, 269), (444, 976), (454, 1313), (459, 75), (468, 443),<br>(469, 289), (475, 67), (484, 1212), (485, 935), (495, 1108), (500, 526),<br>(509, 156), (510, 822), (516, 764), (525, 474), (526, 894), (536, 903),<br>(541, 977), (550, 1550), (551, 1355), (557, 1461), (566, 1417),<br>(567, 853), (577, 698), (582, 1428), (591, 1263), (592, 207), (598, 477),<br>(607, 679), (608, 812), (618, 493), (623, 198), (632, 976), (633, 740),<br>(639, 1174), (648, 1622), (649, 771), (659, 288), (664, 649), (673, 689),<br>(674, 1273), (680, 190), (689, 884), (690, 730), (700, 83), (705, 1100),<br>(714, 402), (715, 125), (721, 887), (730, 146), (731, 689), (741, 1559),<br>(746, 1551), (755, 115), (756, 658), (762, 1584), (771, 1089),<br>(772, 648), (782, 1354), (787, 321), (796, 1509), (797, 1191),<br>(803, 600), (812, 351), (813, 607), (823, 1149), (828, 772), (837, 1222),<br>(838, 43), (844, 1297), (853, 1294), (854, 566), (864, 944), (869, 1223),<br>(878, 935), (879, 576), (885, 313), (894, 556), (895, 525), (905, 739),<br>(910, 1674), (919, 648), (920, 1109), (926, 1010), (935, 1499),<br>(936, 484), (946, 534), (951, 444), (960, 361), (961, 1642), (967, 26),<br>(976, 761), (977, 443), (987, 329), (992, 895), (1001, 74), (1002, 494),<br>(1008, 723), (1017, 23), (1018, 402), (1028, 124), (1033, 1346),<br>(1042, 1468), (1043, 1027), (1049, 1420), (1058, 966), (1059, 361),<br>(1069, 1600), (1074, 116), (1083, 1181), (1084, 1560), (1090, 436),<br>(1099, 228), (1100, 320), (1110, 1395), (1115, 567), (1124, 894),<br>(1125, 412), (1131, 1133), (1140, 1171), (1141, 279), (1151, 1190),<br>(1156, 1018), (1165, 607), (1166, 945), (1172, 149), (1181, 433),<br>(1182, 238), (1192, 985), (1197, 1469), (1206, 320), (1207, 1478),<br>(1213, 846), (1222, 1376), (1223, 197), (1233, 780), (1238, 239),<br>(1247, 33), (1248, 330), (1254, 1543), (1263, 638), (1264, 156),<br>(1274, 575), (1279, 690), (1288, 1427), (1289, 863), (1295, 559),<br>(1304, 1581), (1305, 115), (1315, 370), (1320, 1141), (1329, 1140),<br>(1330, 1396), (1336, 1256), (1345, 843), (1346, 74), (1356, 165),<br>(1361, 1592), (1370, 853), (1371, 248), (1377, 272), (1386, 105),<br>(1387, 33), (1397, 1641), (1402, 362), (1411, 566), (1412, 781),<br>(1418, 969), (1427, 1048), (1428, 1673), (1438, 1436), (1443, 813),<br>(1452, 279), (1453, 1314), (1459, 1666), (1468, 310), (1469, 1632),<br>(1479, 1231), (1484, 1264), (1493, 1673), (1494, 166), (1500, 682),<br>(1509, 1253), (1510, 1591), (1520, 1026), (1525, 34), (1534, 1386),<br>(1535, 699), (1541, 1379), (1550, 515), (1551, 1550), (1561, 821),<br>(1566, 485), (1575, 1099), (1576, 1232), (1582, 395), (1591, 1458),<br>(1592, 1509), (1602, 616), (1607, 936), (1616, 812), (1617, 84),<br>(1623, 1092), (1632, 720), (1633, 1468), (1643, 411), (1648, 1387),<br>(1657, 525), (1658, 617), (1664, 108), (1673, 1663), (1674, 1427) |

| $p$ | $k$ | $B_p$ | classes $(a, b)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|-----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 53  | 1   | 0     | (0, 17), (1, 17), (2, 27), (3, 20), (4, 3), (4, 5), (4, 45), (6, 43), (7, 0),<br>(7, 46), (9, 36), (10, 43), (11, 16), (11, 41), (11, 49), (12, 8), (12, 48),<br>(12, 50), (13, 25), (13, 30), (14, 10), (15, 9), (15, 13), (15, 31), (16, 14),<br>(17, 33), (18, 1), (19, 16), (19, 41), (19, 49), (20, 0), (20, 7), (20, 46),<br>(22, 42), (23, 17), (25, 43), (26, 22), (26, 40), (26, 44), (27, 18),<br>(28, 26), (30, 17), (31, 52), (32, 1), (34, 47), (36, 27), (38, 6), (40, 8),<br>(40, 48), (40, 50), (41, 25), (41, 30), (41, 51), (42, 47), (43, 26),<br>(44, 11), (45, 1), (47, 26), (48, 19), (50, 4), (50, 12), (50, 37), (51, 33),<br>(52, 17)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 53  | 2   | 0     | (7, 749), (13, 846), (60, 219), (66, 422), (113, 2498), (119, 2807),<br>(166, 1968), (172, 2383), (219, 1438), (225, 1959), (272, 908),<br>(278, 1535), (325, 378), (331, 1111), (378, 2657), (384, 687),<br>(431, 2127), (437, 263), (484, 1597), (490, 2648), (537, 1067),<br>(543, 2224), (590, 537), (596, 1800), (643, 7), (649, 1376), (696, 2286),<br>(702, 952), (749, 1756), (755, 528), (802, 1226), (808, 104), (855, 696),<br>(861, 2489), (908, 166), (914, 2065), (961, 2445), (967, 1641),<br>(1014, 1915), (1020, 1217), (1067, 1385), (1073, 793), (1120, 855),<br>(1126, 369), (1173, 325), (1179, 2754), (1226, 2604), (1232, 2330),<br>(1279, 2074), (1285, 1906), (1332, 1544), (1338, 1482), (1385, 1014),<br>(1391, 1058), (1438, 484), (1444, 634), (1491, 2763), (1497, 210),<br>(1544, 2233), (1550, 2595), (1597, 1703), (1603, 2171), (1650, 1173),<br>(1656, 1747), (1703, 643), (1709, 1323), (1756, 113), (1762, 899),<br>(1809, 2392), (1815, 475), (1862, 1862), (1868, 51), (1915, 1332),<br>(1921, 2436), (1968, 802), (1974, 2012), (2021, 272), (2027, 1588),<br>(2074, 2551), (2080, 1164), (2127, 2021), (2133, 740), (2180, 1491),<br>(2186, 316), (2233, 961), (2239, 2701), (2286, 431), (2292, 2277),<br>(2339, 2710), (2345, 1853), (2392, 2180), (2398, 1429), (2445, 1650),<br>(2451, 1005), (2498, 1120), (2504, 581), (2551, 590), (2557, 157),<br>(2604, 60), (2610, 2542), (2657, 2339), (2663, 2118), (2710, 1809),<br>(2716, 1694), (2763, 1279), (2769, 1270) |
| 61  | 1   | 0     | (3, 10), (4, 2), (5, 26), (8, 6), (10, 37), (10, 40), (10, 45), (12, 23),<br>(15, 20), (15, 50), (15, 52), (17, 8), (17, 54), (17, 60), (20, 33), (23, 9),<br>(23, 11), (23, 41), (24, 34), (27, 27), (28, 2), (29, 12), (33, 28), (35, 36),<br>(35, 43), (36, 18), (36, 25), (37, 31), (40, 12), (42, 20), (42, 50),<br>(42, 52), (43, 36), (43, 43), (44, 10), (47, 9), (47, 11), (47, 41), (48, 9),<br>(48, 11), (48, 41), (49, 0), (49, 22), (49, 39), (51, 35), (53, 38), (55, 58),<br>(56, 48), (57, 28), (58, 8), (58, 54), (58, 60)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 61  | 2   | 0     | (9, 1110), (70, 378), (131, 3367), (192, 2635), (253, 1903), (314, 1171),<br>(375, 439), (436, 3428), (497, 2696), (558, 1964), (619, 1232),<br>(680, 500), (741, 3489), (802, 2757), (863, 2025), (924, 1293),<br>(985, 561), (1046, 3550), (1107, 2818), (1168, 2086), (1229, 1354),<br>(1290, 622), (1351, 3611), (1412, 2879), (1473, 2147), (1534, 1415),<br>(1595, 683), (1656, 3672), (1717, 2940), (1778, 2208), (1839, 1476),<br>(1900, 744), (1961, 12), (2022, 3001), (2083, 2269), (2144, 1537),<br>(2205, 805), (2266, 73), (2327, 3062), (2388, 2330), (2449, 1598),<br>(2510, 866), (2571, 134), (2632, 3123), (2693, 2391), (2754, 1659),<br>(2815, 927), (2876, 195), (2937, 3184), (2998, 2452), (3059, 1720),<br>(3120, 988), (3181, 256), (3242, 3245), (3303, 2513), (3364, 1781),<br>(3425, 1049), (3486, 317), (3547, 3306), (3608, 2574), (3669, 1842)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |